

A linear-complexity deep learning network with local-global attention for efficient state-of-health estimation of lithium-ion batteries using optimized multi-source health indicators

Abstract

Efficient and accurate estimation of lithium-ion battery state of health (SOH) is crucial for ensuring safe battery operation and supporting online applications in resource-constrained battery management systems (BMS). However, many existing SOH estimation approaches often rely on health indicators (HIs) with limited stability across cells and resource-consuming model structures. Therefore, this paper proposes a lightweight SOH estimation framework. First, a multi-source health indicator extraction and optimization algorithm is proposed to select HIs that are robust across cells and reduce feature redundancy. Subsequently, a lightweight prediction network, MS-AgentNet, is constructed. The network mainly integrates a Local-Global Fusion Attention (LGFA) module that combines small-kernel depthwise separable convolutions with ReLU² agent attention to capture local degradation features and long-term dependencies, while reducing the quadratic attention complexity of traditional Transformers to linear complexity. Additionally, large-kernel depthwise separable convolutions are used to extract degradation features over longer time scales and work with small-kernel convolutions to fuse multi-scale and multi-channel features with low parameter overhead, enhancing feature diversity. We compared the proposed model with various mainstream deep learning models on four public battery datasets with different chemistries and operating conditions. The experimental results show that MS-AgentNet achieves better overall performance while maintaining low computational and storage overhead, further highlighting the advantages of its lightweight design for resource-constrained BMS.

Keywords: lithium-ion batteries; state-of-health estimation; linear complexity; deep feature fusion; lightweight deep learning

1. Introduction

With high energy density, a long cycle life, and a low self-discharge rate, lithium-ion batteries have become one of the most important electrochemical energy storage technologies, widely used across consumer electronics, electric vehicles, stationary energy storage, and other domains [1–4]. Nevertheless, lithium-ion batteries cannot always remain stable during long-term operation and continue to pose safety risks that cannot be overlooked. Charge-discharge cycling and complex operating conditions induce aging processes such as thickening of the solid electrolyte interphase (SEI) layer, loss of active lithium, and structural degradation of the electrodes [5,6]. These processes, in turn, lead to capacity fade and performance degradation and, in severe cases, trigger failures such as short circuits and thermal runaway [7,8]. Therefore, precise estimation of lithium-ion battery state of health (SOH) is essential for ensuring battery safety, reliability, and operational performance.

Capacity fade is the most direct sign of battery aging, and state of health (SOH) is commonly defined in terms of capacity retention [9]. However, accurately measuring a battery’s maximum available capacity requires full or nearly full charge-discharge cycles, which are impractical for online monitoring [10]. Consequently, indirectly inferring SOH from observable operating signals has become a key approach to online battery health monitoring. Nevertheless, complex operating conditions make it harder to consistently extract degradation information, while the limited computing power and storage space of battery management systems (BMS) constrain model complexity. Together, these factors pose a dual challenge for online SOH estimation: degradation information extraction and the computational efficiency of estimation models.

1.1 Literature review

Recent approaches to SOH estimation can generally be classified as model-based or data-driven [11]. The former include electrochemical models and equivalent circuit models, while the latter include traditional machine learning and deep learning methods.

Model-based approaches simulate the internal electrochemical mechanisms of batteries using mathematical equations or equivalent circuits [12]. Electrochemical models use differential equations to represent electrochemical reactions and degradation processes in batteries. For instance, Li et al. [13] developed an SOH estimation framework based on a single-particle model (SPM), incorporating both chemical degradation and mechanical damage. Although these models are physically interpretable and can achieve high estimation accuracy, solving their complex equations requires extensive computational resources. In addition, parameters such as particle radius and diffusion coefficients are difficult to obtain. These limitations hinder the online application of electrochemical models in resource-constrained BMS [14–16]. In contrast, equivalent circuit models (ECMs) simplify complex electrochemical processes into circuits composed of resistors, capacitors, and other components to simulate battery charge-discharge dynamics. For example, Chen et al. [17] used recursive least squares to identify the parameters of a Thevenin model and estimated SOH based on the relationship between ohmic internal resistance and capacity fade. This simplification reduces computational cost, but it also limits the ability of ECMs to fully capture changes in the internal state of the battery. Compared with electrochemical models, ECMs may achieve lower estimation accuracy. Their accuracy depends on the selected circuit structure and model parameters. Furthermore, ECM parameters are highly sensitive to operating conditions, including temperature and charge-discharge rate, making it difficult for ECMs to maintain robust estimation performance throughout the battery aging process.

In contrast, data-driven approaches avoid the need to develop detailed electrochemical models or extensively analyze electrochemical reactions and battery aging mechanisms [18]. Instead, they train models on historical operating data to map extracted health indicators (HIs) to SOH estimates. Earlier studies primarily employed traditional machine learning models to estimate battery states and predict battery lifetime [19]. For instance, Fei et al. [20] crafted 42 features from charge-discharge data collected over the first 100 cycles. After feature selection, the retained features were fed into six models: elastic net, Gaussian process regression (GPR), support vector machine (SVM), random forest (RF), gradient boosting regression tree (GBRT), and neural network (NN). Their performance in the early prediction of battery lifetime was then compared. Li et al. [21] used a random forest to estimate battery capacity online from partial charging voltage-capacity data. However, when faced with nonlinear and fluctuating data from online monitoring and historical cycling, traditional machine learning models struggle to achieve high predictive performance because of their structural limitations.

With the rapid development of deep learning, researchers have increasingly adopted multilayer neural networks. These networks have strong nonlinear representational capacity and can therefore capture complex battery degradation patterns and improve estimation accuracy [22]. Common deep learning models include convolutional neural networks (CNNs), recurrent neural networks (RNNs), and Transformers. CNNs extract local features using convolutional kernels and reduce feature dimensionality through pooling operations. For example, Qian et al. [23] fed randomly selected charging-curve segments into a one-dimensional CNN for battery capacity estimation. The results showed that the method achieved low capacity estimation errors with randomly selected segments as inputs. Lee et al. [24] transformed capacity fade sequences across cycles into two-dimensional images and employed a two-dimensional CNN for SOH estimation.

However, a single convolutional layer in a traditional CNN has a limited local receptive field, and fusing distant features generally requires multiple convolutional layers [25]. RNNs establish these long-range connections by passing historical information to subsequent time steps through hidden states. RNN variants, including long short-term memory (LSTM) and gated recurrent unit (GRU) networks, further employ gating mechanisms to selectively retain and update historical information to capture long-term dependencies. Researchers have also combined CNNs with recurrent networks to capture local features while retaining historical information. For example, Xu et al. [26] fed CNN-extracted features into an LSTM and used its temporal memory to estimate SOH. Tian et al. [27] further incorporated an attention

mechanism into a CNN-BiLSTM architecture to develop the CNN-BiLSTM-AM prediction model. Zheng et al. [28] employed a CNN-GRU to estimate SOH using segments of voltage, current, and temperature curves obtained during random charging. Nevertheless, the hidden states in these models still require sequential updates across time steps, limiting parallel computation during training and inference.

To address this limitation, Vaswani et al. [29] proposed the Transformer, which avoids recurrence and uses self-attention to directly connect different sequence positions, capturing long-range dependencies while supporting parallel computation. For example, Park and Kim [30] used a Transformer guided by physical priors to estimate long-term battery SOH, while Chen et al. [31] added structures such as dimension-transformation layers to a vision Transformer to adapt it to battery SOH estimation. Researchers have also combined Transformers with convolutional or recurrent networks to enhance prediction accuracy. Gu et al. [32] used a CNN to extract local features and a Transformer to capture long-range dependencies. Jia et al. [33] combined a BiGRU with a Transformer, and their experiments showed good prediction accuracy, robustness, and generalization on the battery data used. Bai and Wang [34] proposed a convolutional Transformer framework that extracts local details from voltage and current signals and uses self-attention to establish global connections among local features. However, many existing methods improve Transformer prediction accuracy by integrating other modules or models, often at the cost of more parameters and higher computational overhead. Meanwhile, the computational bottleneck of standard self-attention remains: it computes pairwise relationships among all sequence positions, with time and memory complexity growing quadratically with sequence length N , namely $O(N^2)$ [35, 36]. These computational and storage demands make it challenging to use such models online in resource-constrained BMS [25].

In addition to the prediction model, health indicator extraction and selection are crucial for accurate SOH estimation. These indicators are mainly extracted from voltage, current, temperature, time, and internal resistance data recorded during battery cycling, as well as curves derived from these data [37, 38]. Some internal-resistance indicators require impedance spectrum measurements, which are time-consuming and require specialized instruments. Although temperature indicators contain battery degradation information, they are also affected by ambient temperature and charge-discharge rate, limiting their stability across operating conditions [39]. Incremental capacity curves reflect changes in voltage plateaus caused by battery aging, but their calculation requires differentiation, which can amplify sampling noise. Smoothing is therefore usually needed to obtain reliable curve features. For example, Wen et al. [40] extracted candidate features such as peak values, peak positions, and peak-shape slopes from smoothed incremental capacity curves and used correlation analysis to select the peak value as the input to a backpropagation (BP) neural network.

In contrast, time-based health indicators do not require differentiation and are simpler to extract. Constant current charge time (CCCT) can be obtained from the difference between the times corresponding to the two endpoints of a selected voltage interval. Richardson et al. [41] used charging-time features from different voltage intervals of constant-current charging curves to estimate battery capacity, while Lin et al. [42] used CCCT as the input to a random forest to estimate battery SOH. Because the correlation between charging-time features and capacity depends on the selected voltage interval, researchers generally need to identify a suitable interval for feature extraction. Tian et al. [43] optimized the charging voltage interval, strengthening the correlation between the extracted health indicator and capacity. In selecting intervals for charging-time features, Li et al. [25] progressively narrowed the voltage window and evaluated candidates using Pearson correlation coefficient (PCC) values from multiple cells in the same group, aiming to extract charging-time features highly correlated with SOH from shorter charging segments.

Different types of operating data and the curves derived from them reflect different aspects of battery degradation. To use this information more fully, some studies construct candidate health indicators from multiple operating signals and further optimize model inputs. For example, Dai et al. [44] extracted health indicators from voltage, current, temperature, incremental capacity, and differential thermal voltammetry curves and calculated statistical features such as the per-cycle means and medians of these indicators. They then compared estimation errors for different combinations of statistical features to determine the inputs for SOH estimation. Lin et al. [45] constructed multiple features from electrical, thermodynamic, and electrochemical perspectives, used principal component analysis to reduce dimension-

ality, and optimized the number of retained dimensions with simulated annealing, thereby reducing redundancy and improving SOH estimation performance.

A single health indicator contains relatively limited degradation information, whereas multi-source features can describe complementary aspects of battery degradation. However, adding more candidate indicators does not necessarily improve estimation accuracy. Indicators with weak representational ability, inconsistent performance across cells, or redundant information may limit the effective use of multi-source information and increase the model’s learning burden. Therefore, a health indicator selection algorithm should be developed to jointly evaluate candidates’ correlations with SOH, performance across cells, and redundancy among indicators. It should retain indicators with stable representational ability and reduce inputs weakly correlated with SOH or containing redundant information, helping the model learn battery degradation patterns more effectively.

A comparison of existing SOH estimation methods is summarized in Table 1. Specifically, we examine whether these methods make full use of degradation information in battery operating data, adopt effective health indicator extraction and selection strategies, balance estimation performance and computational efficiency through targeted model design, and evaluate model complexity. Given differences in battery chemistries and operating conditions, the ability of selected indicators to represent degradation consistently across cells warrants further attention. In addition to estimation performance, model complexity is directly related to practical deployment. In recent years, a notable trend in deep model design has been to increase network depth, expand model size, or introduce more complex structures for feature interaction to enhance representational capacity. These designs help models better capture complex degradation patterns but usually come with higher parameter counts, computational costs, and storage requirements. Therefore, SOH estimation models need to become more compact and efficient while retaining effective representational capacity to meet the real-time estimation requirements of resource-constrained BMS.

Table 1
Comparison of existing SOH estimation methods.

Reference	Source data considered			HI selection method (author-designed)	Estimation model (original design)	Efficiency considered
	Current	Voltage	Temperature			
[46]	✓	✓	✗	PCC (✗)	PINN (✓)	✗
[25]	✗	✓	✗	PCC (✓)	BMSFormer (✓)	✓
[33]	✗	✓	✓	PCC + SCC (✗)	BiGRU-Transformer (✗)	✗
[31]	✓	✓	✗	PCC (✗)	Transformer (✗)	✗
[47]	✓	✓	✗	FOI monotonicity analysis (✓)	Linear regression (✗)	✗
[21]	✗	✓	✗	LC (✗)	Random Forest Regression (✗)	✗
[32]	✓	✓	✓	PCC (✗)	CNN-Transformer (✗)	✓
[45]	✓	✓	✓	SA-PCA (✓)	CNN (✗)	✓
[48]	✗	✓	✗	PCC (✗)	BiLSTM-Transformer (✓)	✗
[49]	✗	✓	✗	Interval selection + feature fusion (✓)	GWO-LSSVM-AdaBoost (✓)	✗
[50]	✗	✓	✓	PCC + input comparison (✗)	AST-LSTM (✓)	✗
[51]	✓	✓	✓	PCC (✗)	LSTM-LAM (✓)	✗
[52]	✓	✓	✓	PCC + SCC (✗)	BAS-ELM (✓)	✗
[42]	✗	✓	✗	CCCT window search (✓)	Random Forest Regression (✗)	✓
[44]	✓	✓	✓	Statistical feature combination optimization (✓)	NGO-Dualkernel-GPR (✓)	✗
[43]	✓	✓	✗	QPSO interval optimization (✓)	LSTM-BP (✗)	✗
This study	✓	✓	✓	Multi-source health indicator extraction and optimization algorithm (✓)	MS-AgentNet (✓)	✓

1.2 Challenges and contributions

Three main challenges facing existing methods are summarized as follows.

(1) **Cross-cell stability of health indicators.** Raw data collected from multiple sensors contain diverse features related to degradation. The relationship between a candidate health indicator and SOH may vary across cells, and an indicator that performs well on one cell may not retain the same representational ability on others. PCC and the Spearman correlation coefficient (SCC) evaluate linear and monotonic relationships, respectively; using only one makes it difficult to assess both aspects. Moreover, even indicators strongly associated with SOH may contain redundant information. Model input selection therefore needs to consider both correlations across cells and redundancy among indicators to reduce its dependence on the performance of a single cell.

(2) **Fusion of local and long-term degradation information.** As cycling progresses, battery capacity generally declines, with fluctuations between neighboring cycles and local capacity recovery. Local features reflect short-term changes, but capturing long-term capacity fade trends also requires connections across cycles. When aggregating information across cycles, local degradation details must be retained so that short-term changes are not weakened. Therefore, the model needs not only to extract degradation features at different time scales but also to make local variations and long-term trends complement each other to improve SOH estimation accuracy.

(3) **Computational complexity limitations.** Many existing deep learning models improve SOH estimation accuracy by increasing network depth or combining different network structures. These approaches increase the number of model parameters and computational operations. Recurrent structures require sequential computation over time steps, while standard self-attention computes pairwise relationships among all sequence positions, with time and memory complexity growing quadratically with sequence length. These costs increase the operating burden on resource-constrained BMS and limit the deployment of online SOH estimation models.

To address these challenges, this paper proposes a lithium-ion battery SOH estimation framework built around MS-AgentNet. The framework addresses health indicator construction and lightweight model design: group-level calibration and selection improve the cross-cell stability of model inputs, while local feature extraction is combined with agent attention to represent local variations and long-term degradation trends with low computational overhead. The main contributions are as follows.

(1) **A multi-source health indicator extraction and optimization algorithm is proposed.** The algorithm first extracts multiple types of candidate health indicators from charge-discharge data and their derived curves. MS-CCCT then calibrates the voltage window for the charging-time feature. The algorithm uses correlation results from the feature-development cells to evaluate each candidate’s association with SOH and its redundancy with other indicators. It then applies PCC and SCC thresholds and redundancy constraints to select the indicators. The same selection rules determine a final HI subset for each dataset, and the selected indicators retain strong linear and monotonic correlations with SOH on other cells in that dataset.

(2) **A lightweight local-global network, MS-AgentNet, is constructed.** To represent both long-term battery degradation trends and short-term capacity fluctuations, small-kernel depthwise separable convolutions are combined with ReLU² agent attention to form a Local-Global Fusion Attention (LGFA) module, integrating local feature extraction and global context modeling with low parameter overhead. The model further introduces large-kernel depthwise separable convolutions, which work with the small-kernel convolutions to extract degradation features at different time scales. With a fixed number of agents and feature dimension, its attention computation reduces the complexity with respect to sequence length from $O(N^2)$ for standard self-attention to $O(N)$. These designs aim to balance SOH estimation accuracy with computational and storage overhead.

(3) **Comprehensive validation is conducted across multiple datasets.** Experiments on multiple public battery datasets with different chemistries, capacities, and charge-discharge protocols, together with module ablation and complexity analysis, provide a comprehensive evaluation of the proposed method in terms of estimation accuracy, computational efficiency, and cross-cell generalization capability. Cross-dataset transfer experiments further examine the model’s transfer performance across different data domains. The results show that the proposed model maintains high SOH estimation accuracy with low computational and storage overhead.

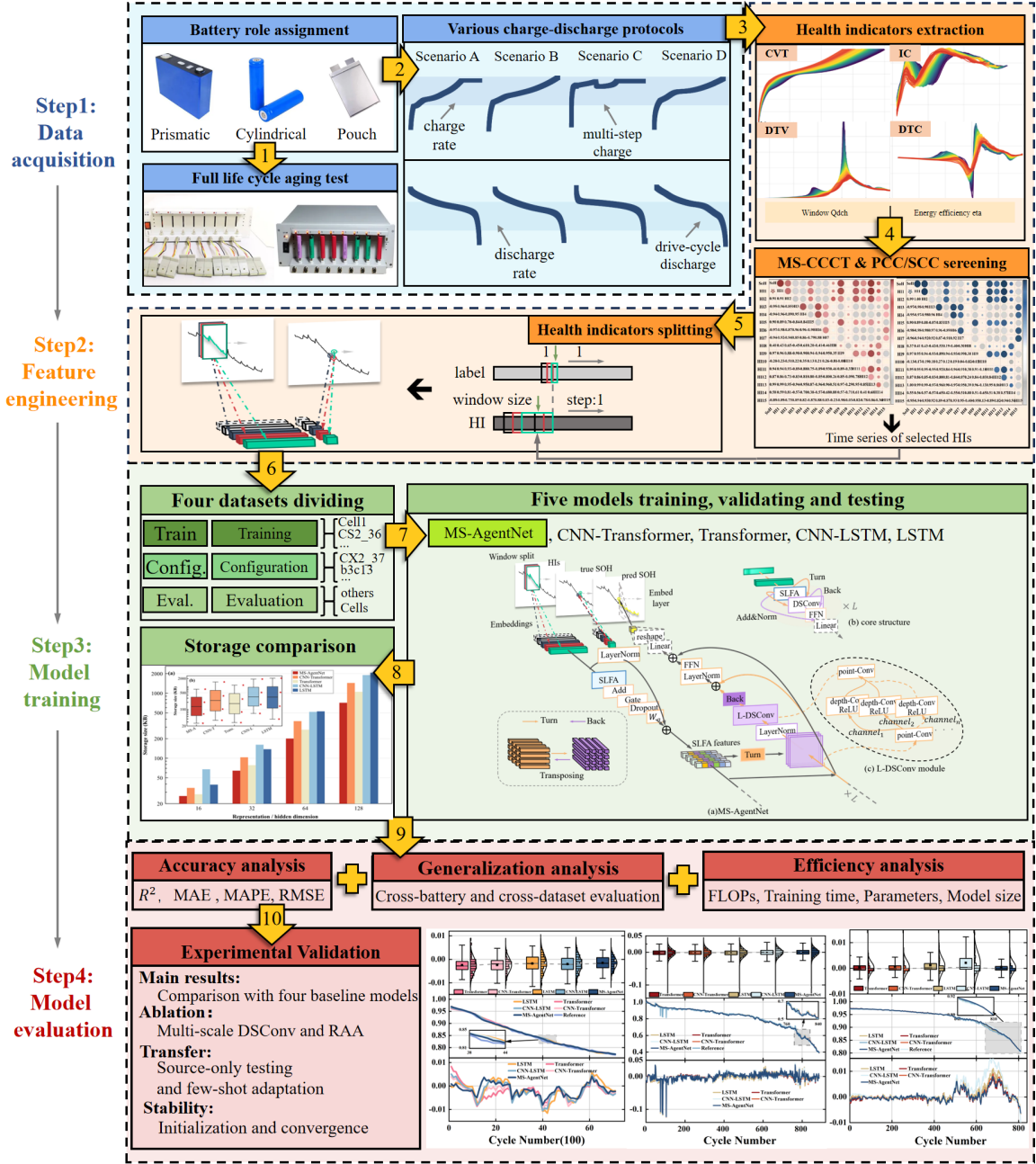


Fig. 1. Flowchart of the developed SOH estimation approach.

2. Methodological framework and data preparation

2.1 Overview of the SOH estimation framework

To use a consistent definition in the following experiments, SOH is calculated as capacity retention, namely the ratio of the current available capacity to the rated capacity [9, 22]:

$$\text{SOH} = \frac{C_{\text{current}}}{C_{\text{rated}}} \times 100\%, \quad (1)$$

where C_{current} and C_{rated} denote the current available capacity and rated capacity of the battery, respectively.

The proposed SOH estimation framework is shown in Fig. 1.

(1) **Data acquisition.** Public battery aging data covering different battery chemistries and operating conditions are selected to obtain cycle records of voltage, current, temperature, time, and capacity during charging and discharging.

(2) **Systematic feature engineering.** The proposed multi-source health indicator extraction and optimization algorithm constructs a pool of multi-source candidate HIs from these data and determines the model inputs through

group-level calibration and selection. A sliding window is then used to segment the HI time series. Each window is paired with the SOH of the next cycle to form an input sample and its label.

(3) **Model training.** Samples of the selected HIs are fed into MS-AgentNet. Model parameters are learned on the training cell, while the validation cell is used for validation and comparison to select the best-performing hyperparameter configuration. The selected model is then directly tested on the remaining cells in the corresponding dataset.

(4) **Model evaluation.** The SOH estimation accuracy, cross-cell generalization, and computational and storage overhead of the different models are evaluated. Ablation studies are further conducted to analyze the roles of key modules, and cross-dataset transfer experiments are used to examine cross-domain adaptation.

2.2 Typical battery datasets

Four public battery aging datasets, Oxford, CALCE CS2, CALCE CX2, and MIT/Severson, are selected to examine the generalization capability of the proposed method across different battery systems and operating conditions. These datasets include pouch, prismatic, and cylindrical cells and differ considerably in battery chemistries, nominal capacities, and operating conditions, providing diverse experimental data for model evaluation under different battery degradation scenarios. Their main characteristics are summarized in Table 2. Fig. 2(a)–(d) shows the capacity degradation curves of the selected cells, and Fig. 2(e)–(h) presents the charging voltage curves of a representative cell from each dataset at different SOH levels. Details of each dataset are given below.

(1) Oxford dataset

The Oxford dataset is provided by the Battery Intelligence Laboratory at the University of Oxford and contains aging data from 8 Kokam SLPB533459H4 lithium-ion pouch cells, each with a nominal capacity of 0.74 Ah [41]. The cathode is a blend of lithium nickel cobalt oxide (NCO) and lithium cobalt oxide (LCO), and the anode is graphite [53]. The aging tests were conducted at a constant temperature of 40 °C. The cells were charged to 4.2 V using a constant-current–constant-voltage protocol and then discharged following a dynamic current profile derived from the urban ARTEMIS drive cycle until the voltage reached 2.7 V [54]. Aging records from cycling tests on all 8 cells, Cell1 – Cell8, are included in the experiments.

(2) CALCE dataset

The CALCE dataset is provided by the Center for Advanced Life Cycle Engineering at the University of Maryland and contains aging data from two groups of prismatic lithium-ion cells, CS2 and CX2 [54,55]. Both groups use lithium cobalt oxide (LCO) as the cathode material, with nominal capacities of 1.1 Ah for CS2 and 1.35 Ah for CX2. All aging tests were conducted at room temperature, and the cells were charged to 4.2 V using a constant-current–constant-voltage protocol. Both CS2 and CX2 were discharged at a constant current of 1C to 2.7 V. Aging records from cycling tests on 6 cells, CS2_36–CS2_38 and CX2_36–CX2_38, are included in the experiments.

(3) MIT/Severson dataset

The MIT/Severson dataset was jointly released by the Massachusetts Institute of Technology, Stanford University, and Toyota Research Institute [56]. It uses A123 APR18650M1A 18650 cylindrical lithium-ion cells, each with a nominal capacity of approximately 1.1 Ah, a lithium iron phosphate (LFP) cathode, and a graphite anode. The aging tests were conducted at a constant temperature of 30 °C. Three cells, b3c8, b3c13, and b3c29, are included in the experiments. All three cells followed two-step fast-charging protocols: 5.3C (54% SOC)–4C for b3c8 and 5.6C (36% SOC)–4.3C for b3c13 and b3c29. After reaching 80% SOC, the cells were charged at a constant current of 1C to 3.6 V, followed by constant-voltage charging. Discharging was performed at a constant current of 4C with a cutoff voltage of 2.0 V.

2.3 Feature engineering

This section presents the multi-source health indicator extraction and optimization algorithm for constructing the inputs to the SOH estimation model. The algorithm first builds a pool of multi-source candidate HIs from charging and discharging data, then optimizes the model inputs through group-level calibration and selection. Specifically, MS-CCCT calibrates the CCCT voltage window, and PCC/SCC results on the feature-development cell set are used for

Table 2

Description of the four battery datasets.

Data Sources	Oxford [53]	CALCE CS2 [54, 55]	CALCE CX2 [54, 55]	MIT/Severson [56]
Manufacturer	Kokam	LG Chem	LG Chem	A123 Systems
Cell types	Pouch	Prismatic	Prismatic	18650 cylindrical
Cathode material	NCO-LCO	LCO	LCO	LFP
Rated capacity (Ah)	0.74	1.10	1.35	1.10
Capacity failure threshold	70%	0.84 Ah (76.4%)	1.08 Ah (80%)	80%
Charge protocol (rate)	CC (2C)	CC-CV (0.5C, 4.2 V)	CC-CV (0.5C, 4.2 V)	Two-step fast charge + CC-CV (1C, 3.6 V)
Discharge protocol (rate)	Dynamic current profile	CC (1C)	CC (1C)	CC (4C)
Cut-off voltage (V)	charge to 4.2 discharge to 2.7	charge to 4.2 discharge to 2.7	charge to 4.2 discharge to 2.7	charge to 3.6 discharge to 2.0
Cut-off current (A)	—	charge to 0.05	charge to 0.05	charge to 0.022 (C/50)
Experiment temperature (°C)	40	24	24	30

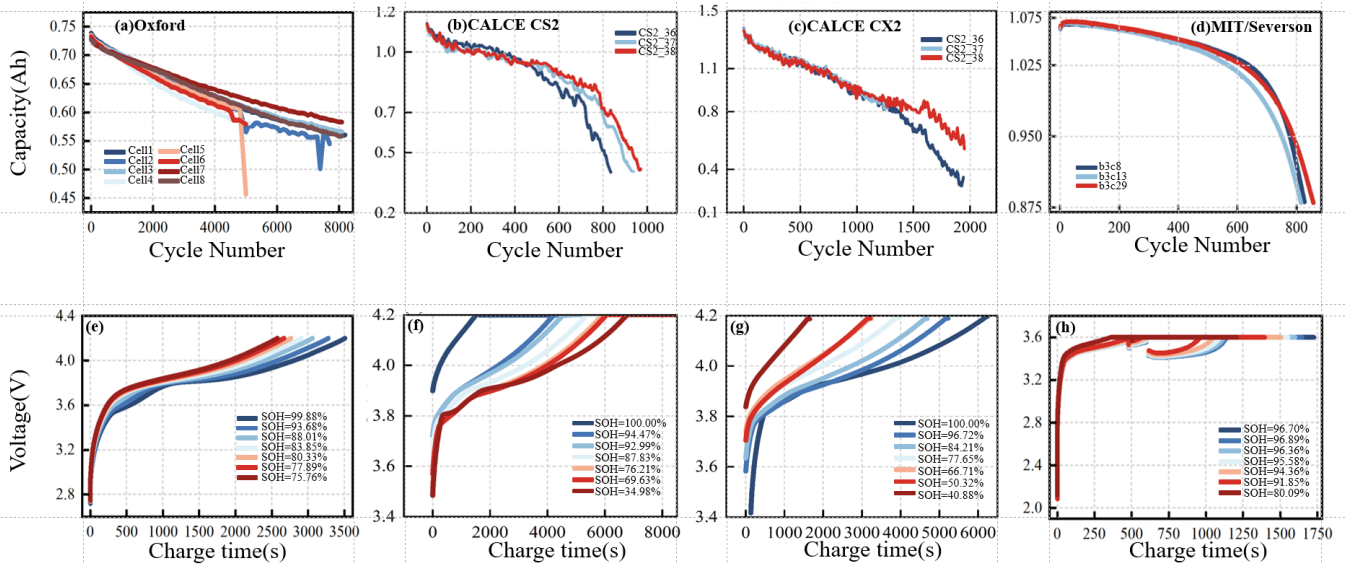


Fig. 2. Four battery datasets: (a)–(d) capacity degradation curves of Oxford, CALCE CS2, CALCE CX2, and MIT/Severson cells, respectively; (e)–(h) charging voltage curves of the four datasets in the same order.

dual-threshold admission and redundancy removal to determine the inputs for each dataset.

2.3.1 Health indicator extraction

Charging voltage–time (CVT), incremental capacity (IC), differential temperature–voltage (DTV), and differential temperature–capacity (DTC) curves are constructed from voltage, current, temperature, time, and capacity data collected during battery cycling, as shown in Fig. 3(a)–(d). Features describing duration, amplitude, characteristic position, and distribution width are extracted based on how these curves change with aging. Discharge capacity within a voltage window and charge-discharge energy efficiency are also calculated, forming a pool of 15 candidate HIs [39, 42, 57, 58]. Their definitions and categories are listed in Table 3.

The charging voltage–time (CVT) curve records changes in terminal voltage over time during constant-current charging and can be obtained directly from voltage and time samples. As cycling progresses, the CVT curve gradually shifts along the time axis, as shown in Fig. 3(a). This shift appears as a continuous change in charge duration within a fixed voltage interval as the battery ages. Increased polarization and capacity fade caused by battery aging jointly change this charging process [39, 42]. Accordingly, CCCT is defined as:

$$CCCT(V_1, V_2) = t(V_2) - t(V_1), \quad (2)$$

where $t(V_1)$ and $t(V_2)$ are the times when the charging voltage reaches V_1 and V_2 , respectively. The CCCT voltage interval is calibrated by MS-CCCT, as described in Section 2.3.2, and the resulting charging-time feature is denoted as HI1.

In addition to charging-time features, changes in capacity with voltage also contain battery degradation information. The incremental capacity (IC) curve is a well-established tool for analyzing electrochemical degradation in batteries. Differentiating capacity with respect to voltage converts relatively flat charging voltage plateaus into more distinct peaks whose positions and shapes change with battery aging [39, 40]. IC is expressed as:

$$IC = \frac{dQ}{dV} \approx \frac{Q_{k+N} - Q_k}{V_{k+N} - V_k}, \quad (3)$$

where V_k and Q_k are the terminal voltage and charge capacity at the k th sampling point, respectively, and N is the interval between the two sampling points. Across cycles, the peak value, characteristic position, and distribution width of the IC curve change with aging, as shown in Fig. 3(b). The peak value, voltage corresponding to the peak, and full width at half maximum of the IC curve are therefore extracted and defined as HI2, HI3, and HI4, respectively.

Temperature is another important measure of the internal state of a battery. During constant-current charging, the battery surface temperature usually changes slowly, making small aging-related differences in thermal response difficult to identify directly from the raw temperature curve. To identify these differences, Wu et al. [58] proposed differential temperature–voltage (DTV) analysis. By calculating the rate of change in battery surface temperature with respect to voltage during charging, this method maps the thermal response to the voltage domain, producing a curve with peaks and valleys that change with aging [39]. DTV is defined as:

$$DTV = \frac{dT}{dV} \approx \frac{T_{k+N} - T_k}{V_{k+N} - V_k}, \quad (4)$$

where V_k and T_k are the terminal voltage and battery surface temperature at the k th sampling point, respectively, and N is the interval between the two sampling points. To reduce the effect of noise introduced by differencing, the DTV curve is smoothed using a moving average with a window size of 200 sampling points. Across cycles, the peak and valley values and characteristic positions of the DTV curve change with aging, as shown in Fig. 3(c). The peak value (HI5), voltage corresponding to the peak (HI6), valley value (HI7), voltage corresponding to the valley (HI8), and peak-to-valley difference (HI9) are extracted to describe the amplitude, characteristic positions, and range of variation of the thermal response [39].

While DTV characterizes the thermal response in the voltage domain, DTC characterizes it in the charge capacity domain. With charge capacity as the independent variable, DTC describes how the rate of temperature change evolves with capacity and is defined as follows [39]:

$$DTC = \frac{dT}{dQ} \approx \frac{T_{k+N} - T_k}{Q_{k+N} - Q_k}, \quad (5)$$

where T_k and Q_k are the battery surface temperature and charge capacity at the k th sampling point, and N is the interval between the two sampling points. The DTC curve is also smoothed using a moving average with a window size of 200 sampling points. The DTC curves in Fig. 3(d) likewise show changes in peak value, characteristic position, and distribution width across cycles. Accordingly, the peak value, capacity corresponding to the peak, and full width at half maximum are extracted and defined as HI10, HI11, and HI12.

Discharge capacity within a voltage window is obtained by integrating the discharge current over a specified voltage interval and is defined as:

$$Q_{\text{dch}}(V_h, V_l) = \frac{1}{3600} \int_{\{t: V_l \leq V_{\text{dch}}(t) \leq V_h\}} |I_{\text{dch}}(t)| dt, \quad (6)$$

where $V_{\text{dch}}(t)$ and $I_{\text{dch}}(t)$ are the discharge voltage and current, respectively, and integration is performed over the discharge time segments in which the voltage lies within $[V_l, V_h]$. Time is measured in s, and Q_{dch} is measured in Ah. Two candidate indicators of discharge capacity within a voltage window are further constructed by calculating the charge released as the terminal voltage decreases from 3.80 V to 3.40 V and from 3.20 V to 3.00 V. These indicators are denoted as HI13 and HI14, respectively [25, 57].

Energy efficiency reflects the energy conversion characteristics of a battery during charging and discharging and is defined as the ratio of discharge energy to charge energy in a single cycle [59]:

$$\eta = \frac{E_{\text{dch}}}{E_{\text{ch}}} = \frac{\int_0^{t_{\text{dch}}} V_{\text{dch}}(t) |I_{\text{dch}}(t)| dt}{\int_0^{t_{\text{ch}}} V_{\text{ch}}(t) |I_{\text{ch}}(t)| dt}, \quad (7)$$

where $V_{\text{ch}}(t)$, $|I_{\text{ch}}(t)|$, and t_{ch} denote the terminal voltage, current magnitude, and duration of charging, respectively; $V_{\text{dch}}(t)$, $|I_{\text{dch}}(t)|$, and t_{dch} denote those of discharging. The full charging and discharging phases of each cycle are used. This energy efficiency is denoted as HI15 (η).

Table 3

Definitions of the candidate HIs.

No.	Category	Health indicator
HI1	CVT	Constant current charge time within the voltage interval calibrated by MS-CCCT
HI2	IC	Peak value of the IC curve
HI3	IC	Voltage corresponding to the IC peak
HI4	IC	Full width at half maximum of the IC peak
HI5	DTV	Peak value of the DTV curve
HI6	DTV	Voltage corresponding to the DTV peak
HI7	DTV	Valley value of the DTV curve
HI8	DTV	Voltage corresponding to the DTV valley
HI9	DTV	Peak-to-valley difference of the DTV curve
HI10	DTC	Peak value of the DTC curve
HI11	DTC	Capacity corresponding to the DTC peak
HI12	DTC	Full width at half maximum of the DTC peak
HI13	Discharge capacity within a voltage window	Discharge capacity within 3.80–3.40 V
HI14	Discharge capacity within a voltage window	Discharge capacity within 3.20–3.00 V
HI15	Energy efficiency	Ratio of discharge energy to charge energy in a single cycle

2.3.2 Group-level dual-correlation MS-CCCT

Constant current charge time can reflect changes in battery capacity with cycling, but its correlation with SOH depends on the voltage interval used for extraction. Some studies directly extract charging-time features from predefined local voltage intervals [41, 42]. However, degradation-sensitive intervals differ across datasets, and a fixed window used for one dataset is difficult to apply directly to others. To improve the correlation between CCCT and SOH and its stability across cells within the same dataset, this study proposes a group-level dual-correlation multi-scale search method (MS-CCCT). Two cells from the same dataset form the feature-development set $\mathcal{B}$. Correlation results within this set are used to construct a group-level robust score, which determines the extraction window for HI1.

Most existing CCCT window optimization methods use PCC to evaluate candidate intervals [25, 43]. PCC is a common statistical measure of the strength of the linear correlation between CCCT and SOH. However, it is sensitive only to linear changes and is insufficient on its own to characterize monotonic nonlinear relationships. SCC specifically measures monotonic relationships between variables and is suitable for characterizing such nonlinear features. Both PCC and SCC are used to evaluate candidate windows, accounting for the linear correlation and monotonic relationship between CCCT and SOH [19, 45, 60]. The PCC $\gamma_{i,j}$ and SCC $\rho_{i,j}$ of the i th candidate window on the j th cell in $\mathcal{B}$ are defined as

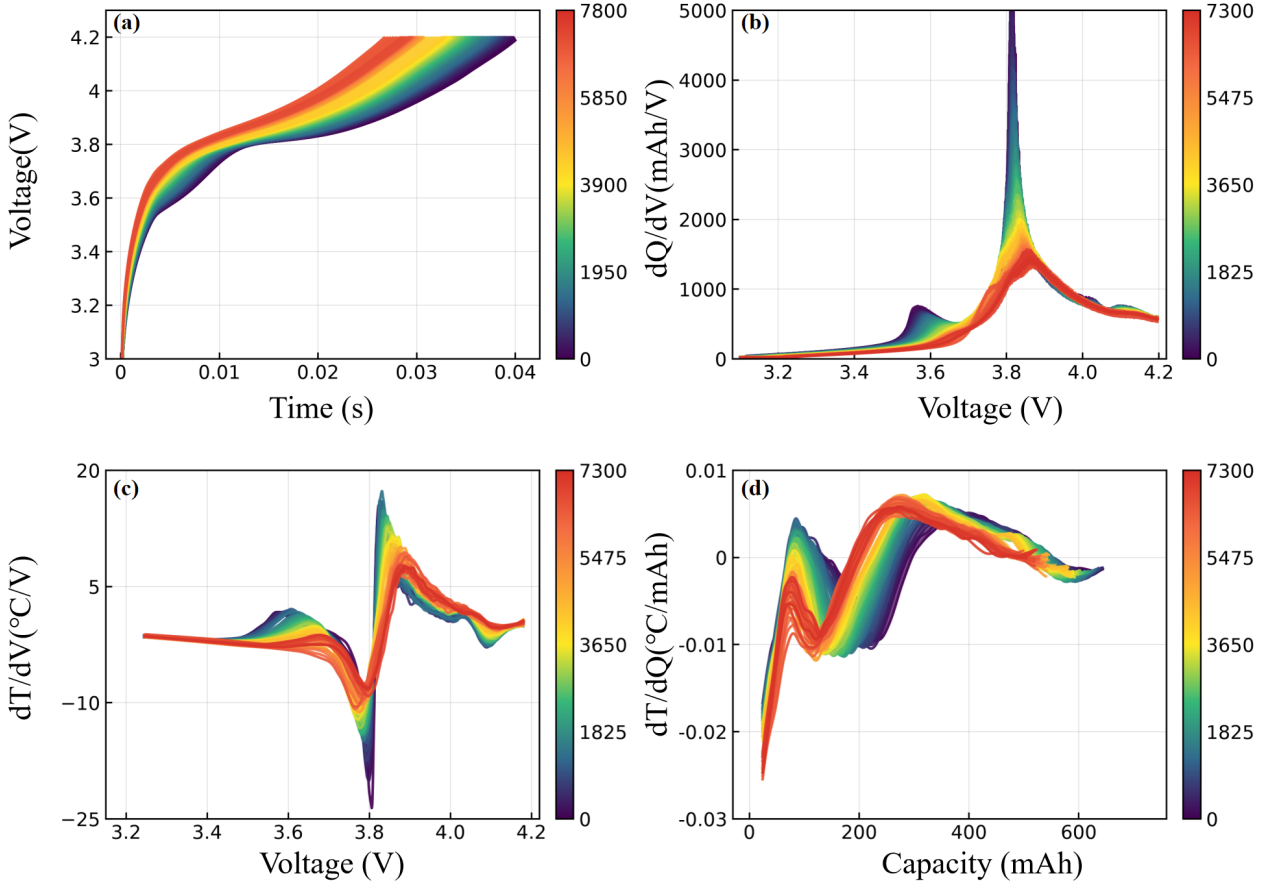


Fig. 3. Charge-side characteristic curves used for HI construction.

$$\gamma_{i,j} = \frac{\sum_{k=1}^{n_j} (f_{i,j,k} - \bar{f}_{i,j})(y_{j,k} - \bar{y}_j)}{\sqrt{\sum_{k=1}^{n_j} (f_{i,j,k} - \bar{f}_{i,j})^2 \sum_{k=1}^{n_j} (y_{j,k} - \bar{y}_j)^2}}, \quad (8)$$

$$\rho_{i,j} = \frac{\sum_{k=1}^{n_j} (R(f_{i,j,k}) - \bar{R}(f_{i,j}))(R(y_{j,k}) - \bar{R}(y_j))}{\sqrt{\sum_{k=1}^{n_j} (R(f_{i,j,k}) - \bar{R}(f_{i,j}))^2 \sum_{k=1}^{n_j} (R(y_{j,k}) - \bar{R}(y_j))^2}}, \quad (9)$$

where $f_{i,j,k}$ is the CCCT extracted from the i th candidate window at the k th cycle of the j th cell, $y_{j,k}$ is the SOH of the corresponding cycle, and n_j is the number of valid cycles for that cell. $\bar{f}_{i,j}$ and $\bar{y}_j$ are the mean CCCT and SOH, respectively, and $R(\cdot)$ denotes the ranking operation. Both PCC and SCC range from -1 to 1 , with absolute values closer to 1 indicating stronger correlations.

To account for the correlation performance of each candidate window on both cells, its minimum absolute PCC and minimum absolute SCC within $\mathcal{B}$ are calculated:

$$mPCC_i = \min_{j \in \mathcal{B}} |\gamma_{i,j}|, \quad mSCC_i = \min_{j \in \mathcal{B}} |\rho_{i,j}|.$$

The smaller of these two values is then used as the group-level robust score of the i th candidate window:

$$S_i = \min(mPCC_i, mSCC_i). \quad (10)$$

This score is determined by the lowest correlation across the two cells, preventing a locally high correlation on a single cell from dominating window selection and allowing the selected window to maintain strong correlations on both cells.

MS-CCCT uses a three-stage coarse-to-fine search. For the Oxford dataset, the initial search range is 3.50–4.20 V. Stage R1 uses a window width of 0.40 V and a step size of 0.20 V for coarse localization, and the window with the highest robust score becomes the search region for the next stage. Stage R2 searches this region using a window width of 0.20 V and a step size of 0.05 V. Stage R3 further refines the search using a window width of 0.10 V and a step size of 0.05 V. If the robust score obtained in R3 exceeds that in R2, the R3 window is selected; otherwise, the R2 result is retained.

For the Oxford dataset, the 3.55–3.75 V window is finally selected with the highest robust score of 0.994447, and its constant current charge time series is used as HI1.

2.3.3 Health indicator selection

HI selection is a key step in lithium-ion battery SOH estimation [39, 45], as the quality of the selected indicators directly affects the final estimation accuracy. Following HI1 window calibration, 15 features based on constant current charge time, IC, DTV, DTC, discharge capacity within a voltage window, and energy efficiency are considered as candidate HIs. Using the same group-level correlation evaluation, PCC and SCC are calculated both between each candidate HI and SOH and between different candidate HIs.

The correlation results for Oxford Cell1 and Cell2 are shown in Fig. 4, including both PCC and SCC between candidate HIs and SOH and among the candidate HIs. The lower triangular part of each matrix lists the correlation coefficients, while the upper triangular part represents correlation strength through circle size and color intensity. Larger, darker circles indicate stronger correlations between the corresponding variables.

The same candidate HI does not show identical correlation performance on the two cells. For HI5 (DTV peak value), the absolute PCC and SCC are 0.949 and 0.966 on Cell1 but decrease to 0.898 and 0.902 on Cell2. In contrast, HI1 maintains high correlations on both cells. Based on the results for Cell1 alone, HI5 would appear to be strongly correlated, but this result does not reflect its correlation level on Cell2. Thus, a correlation ranking based on a single cell cannot fully represent the performance of candidate HIs on other cells.

Existing studies usually select HIs based on the magnitude of their correlation coefficients. When the correlation distributions of candidate HIs change with battery chemistry, operating temperature, or other conditions, the selection results can be sensitive to the data conditions. An indicator that performs well under one condition may become less applicable to other cells or operating conditions. To improve the robustness of HI selection to differences between individual cells and operating conditions, a selection strategy that accounts for the characteristics of correlation distributions is needed.

Accordingly, this study develops a group-level HI selection method. It first considers PCC and SCC on both cells in the feature-development set and retains only candidate HIs that meet all correlation requirements. The correlation matrices also show high correlations between some candidate HIs, indicating redundant information in the candidate pool. The retained HIs are ranked by correlation strength, and features highly correlated with already selected HIs are further removed to reduce redundancy while preserving feature diversity. The specific selection criteria, thresholds, and complete procedure are given in Table 4.

After correlation-based admission and redundancy removal, HI1 is retained for the Oxford dataset. Its absolute PCC values on Cell1 and Cell2 are 0.998759 and 0.997322, and its absolute SCC values are 0.998710 and 0.994447, respectively. All four coefficients are close to 1 and differ only slightly between the two cells, indicating that HI1 is highly sensitive to SOH changes during degradation and has good cross-cell consistency.

HI selection is performed only during feature development. Once the indicators are determined, their definitions and calculation parameters remain fixed and are directly applied to feature extraction for the other cells in the corresponding dataset without correlation-based selection, threshold determination, or HI reselection.

With its definition fixed, HI1 achieves absolute PCC values of 0.997874–0.999337 on Cell3–Cell8. Comparisons with existing HIs are presented in Table 5. Its correlations exceed the reported values for the listed sample entropy,

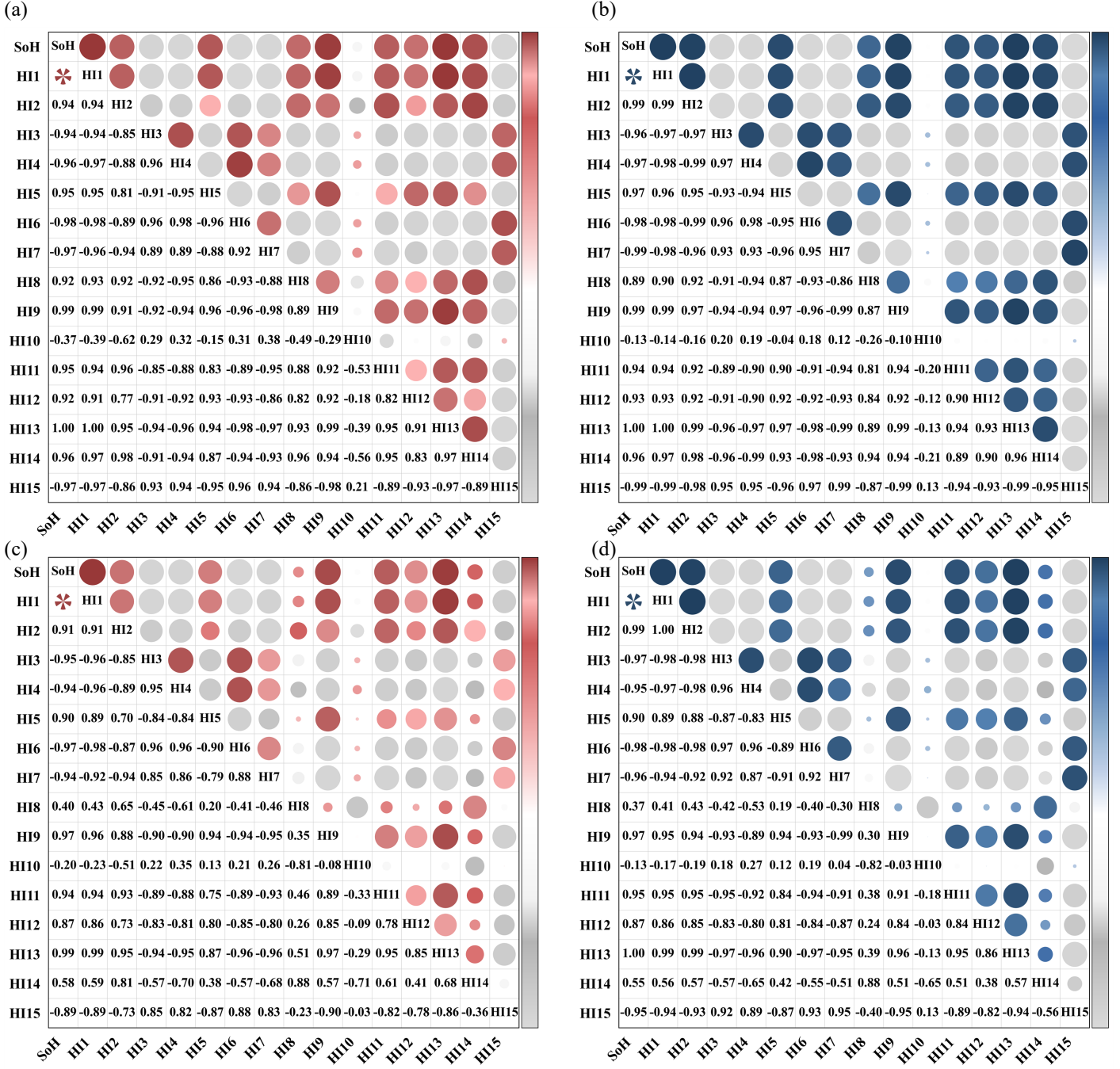


Fig. 4. Correlation matrices of candidate HIs and SOH on the Oxford dataset: (a) Cell1, PCC; (b) Cell1, SCC; (c) Cell2, PCC; (d) Cell2, SCC.

DTV/IC features, and charging voltage curve slopes on the corresponding cells, showing that HI1 maintains high linear correlations on the remaining six cells.

Table 4

Group-level health indicator selection steps.

Procedure: Group-level health indicator selection

1. Calculate the PCC and SCC of each candidate HI on the cells in the feature-development set $\mathcal{B}$.
2. Obtain $mPCC_i = \min_{j \in \mathcal{B}} |\gamma_{i,j}|$ and $mSCC_i = \min_{j \in \mathcal{B}} |\rho_{i,j}|$ for each candidate HI f_i .
3. Retain the HIs satisfying $mPCC_i \geq \theta_{\text{lin}}$ and $mSCC_i \geq \theta_{\text{mono}}$, where $\theta_{\text{lin}} = 0.92$ and $\theta_{\text{mono}} = 0.96$, and sort them in descending order according to $R_i = \max(mPCC_i, mSCC_i)$.
4. Initialize the selected HI set $\mathcal{S} = \emptyset$.
5. For each retained candidate HI f_i :
 - compare f_i with each HI already contained in $\mathcal{S}$;
 - if their group-average absolute PCC and SCC both reach $\theta_{\text{co}} = 0.95$, remove f_i ;
 - otherwise, add f_i to $\mathcal{S}$.
6. Output the final HI subset $\mathcal{S}$.

Table 5

Absolute PCC comparison of different HIs on the Oxford dataset.

Cell	HI	Sample entropy [61]	PP_{DTV} [62]	P_{IC} [62]	DTV peak position FV_6 [63]	Voltage slope [64]
Cell3	0.999052	0.9876	0.9393	0.9729	0.964	0.9980
Cell4	0.998933	0.9885	0.9102	0.9817	0.972	—
Cell5	0.997874	0.9133	0.9574	0.9034	—	—
Cell6	0.998257	0.9848	0.9299	0.9788	—	—
Cell7	0.999337	0.9896	0.9676	0.9748	0.983	0.9982
Cell8	0.999259	0.9908	0.9654	0.9713	0.981	0.9979

3. The proposed MS-AgentNet model

A lightweight Multi-Scale Agent Network (MS-AgentNet) is proposed to represent both local variations and long-term trends in battery degradation while reducing the computational overhead of sequence modeling. The name MS refers to the multi-scale convolutional design comprising small-kernel DSConv-S and large-kernel DSConv-L, which extracts degradation features at different scales with low parameter overhead. Local-Global Fusion Attention (LGFA) combines local convolutional enhancement with efficient attention to connect local degradation information with context across cycles. The following subsections introduce the overall architecture and workflow of MS-AgentNet, describe the designed multi-scale depthwise separable convolution modules, and detail the LGFA module.

3.1 Architecture overview of MS-AgentNet

The overall architecture of MS-AgentNet is shown in Fig. 5. Health indicator (HI) sequences are divided into sample segments using a sliding window, mapped to a d -dimensional feature space through a linear embedding layer, and fed into an MS-AgentNet Block. The Block output passes through layer normalization, flattening, and a linear readout layer to produce the SOH estimate. During training, the true SOH of the cycle immediately following each input window is used as the supervision label.

Mathematically, let the input to the l th MS-AgentNet Block be $\mathbf{X}_l \in \mathbb{R}^{B \times N \times d}$, where B is the batch size, N is the input sequence length, and d is the feature embedding dimension. The Block transforms the features in three steps:

Step 1: Local-global feature fusion. The input $\mathbf{X}_l$ is processed by LGFA, which combines DSConv-S for local degradation feature extraction with ReLU² Agent Attention (RAA) for global context modeling within the input window:

$$\mathbf{X}'_l = \text{LGFA}(\mathbf{X}_l). \quad (11)$$

Step 2: Feature refinement over longer time scales. The LGFA output $\mathbf{X}'_l$ passes through layer normalization and then DSConv-L to extract degradation features over longer time scales. The DSConv-L output is scaled by W_l and added to $\mathbf{X}'_l$ through a residual connection:

$$\mathbf{X}_l'' = \mathbf{X}_l' + W_l \text{DSConv-L}(\text{LN}(\mathbf{X}_l')). \quad (12)$$

Step 3: Nonlinear mapping. The features $\mathbf{X}_l''$ are processed by LN_0 and then a feedforward neural network (FFN). The result is added to $\mathbf{X}_l''$ through a residual connection to produce the Block output $\mathbf{Y}_l$:

$$\mathbf{Y}_l = \mathbf{X}_l'' + \text{FFN}(\text{LN}_0(\mathbf{X}_l'')). \quad (13)$$

The output $\mathbf{Y}_l$ is used as the input to the next Block, i.e., $\mathbf{X}_{l+1} = \mathbf{Y}_l$. Here, W_l is a learnable scaling factor. Both LN and LN_0 denote layer normalization, but LN_0 uses no additional scale or bias parameters.

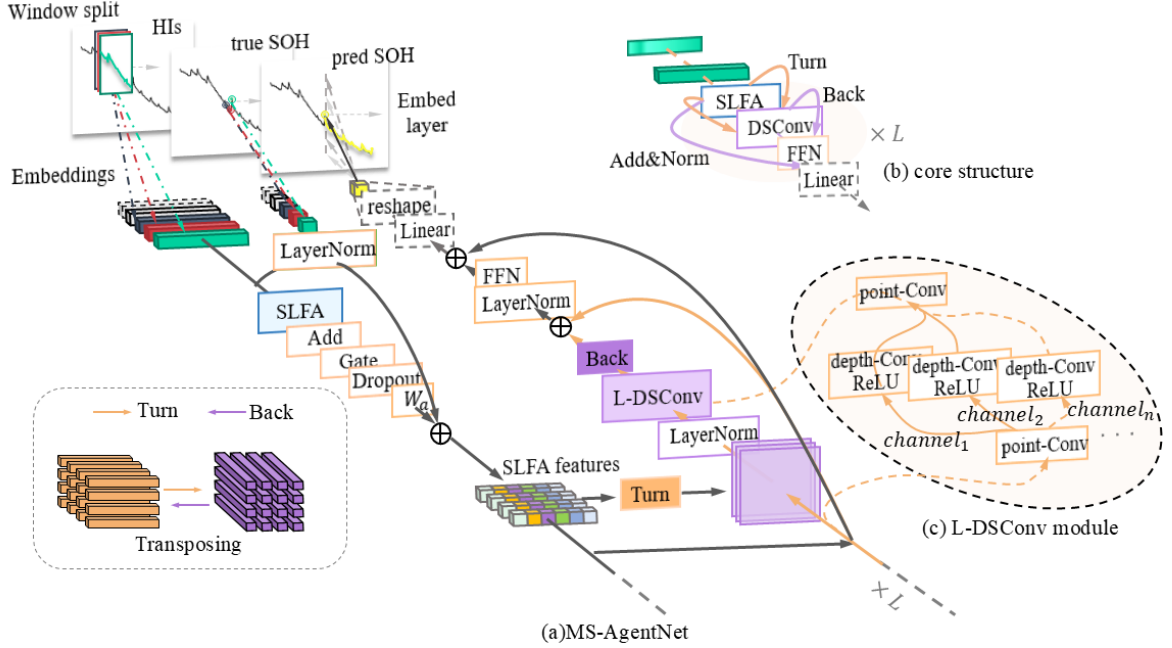


Fig. 5. Framework of MS-AgentNet. (a) Overall architecture; (b) core structure; (c) DSConv-L module.

3.2 The designed multi-scale depthwise separable convolution modules

To balance feature extraction capability and computational efficiency, small- and large-kernel depthwise separable convolutions are used to capture multi-scale local dependencies, enhancing feature diversity while limiting parameter overhead. This section first introduces the computational characteristics of standard and depthwise separable convolutions, followed by the configurations and roles of DSConv-S and DSConv-L.

3.2.1 Basic DSConv structure and computational cost comparison

Convolution extracts local information from time series through sliding kernels. Standard convolution operates across all input channels, with each kernel producing one output feature map corresponding to one output channel. As the numbers of input and output channels and the kernel size increase, the parameter count and computational overhead grow rapidly, substantially increasing the computational load and training time. A larger parameter count may also increase the risk of overfitting on small-sample battery datasets. The computational cost of standard convolution can be expressed as:

$$C_{\text{Conv}} = k \times k \times C_{\text{in}} \times C_{\text{out}} \times D_F \times D_F. \quad (14)$$

where k , C_{in} , C_{out} , and D_F denote the kernel size, number of input channels, number of output channels, and feature map size, respectively.

Unlike standard convolution, depthwise separable convolution (DSConv) decomposes convolution into two stages: depthwise convolution and pointwise convolution [65]. Depthwise convolution applies a separate kernel to each input channel to extract local features within that channel. A 1×1 pointwise convolution then integrates features

across channels and adjusts the number of channels to C_{out} . By separating within-channel feature extraction from cross-channel feature fusion, DSConv reduces the dense cross-channel operations required by standard convolution. Its computational cost can be expressed as:

$$C_{\text{DSConv}} = k \times k \times C_{\text{in}} \times D_F \times D_F + C_{\text{in}} \times C_{\text{out}} \times D_F \times D_F. \quad (15)$$

The ratio of the computational costs of the two types of convolution is:

$$\begin{aligned} \frac{C_{\text{DSConv}}}{C_{\text{Conv}}} &= \frac{k \times k \times C_{\text{in}} \times D_F \times D_F + C_{\text{in}} \times C_{\text{out}} \times D_F \times D_F}{k \times k \times C_{\text{in}} \times C_{\text{out}} \times D_F \times D_F} \\ &= \frac{1}{C_{\text{out}}} + \frac{1}{k^2}. \end{aligned} \quad (16)$$

Equation (16) shows that, with the same input and output channel configurations, DSConv has a lower computational cost than standard convolution, providing the basis for the small- and large-kernel modules described below.

3.2.2 DSConv-S: Dual-role local enhancement

Small-kernel DSConv-S uses a compact 1×5 depthwise convolution and is embedded in LGFA before the global interactions in RAA. It serves two roles: input enhancement and local feature preservation:

1. **Input enhancement.** To extract fine-grained degradation information from health indicator sequences, DSConv-S models local relationships between the feature representations of neighboring cycles to form a locally enhanced representation $\mathbf{X}_S$. Depthwise convolution extracts local dependencies within each channel, while pointwise convolution integrates features across channels. RAA then constructs queries, keys, and values from $\mathbf{X}_S$ and establishes relationships across cycles through agent aggregation and broadcasting, allowing context aggregation to use the local variation information extracted by convolution.

2. **Local feature preservation.** Alongside information interactions across cycles, $\mathbf{X}_S$ passes through layer normalization and is sent directly through the local branch to the fusion stage, where it is added to the RAA branch output. This path combines the locally enhanced health indicator representation with the global context established by RAA within the input window to form a joint local-global representation for SOH estimation.

DSConv-S uses a channel expansion factor of two. The input features are $\mathbf{X} \in \mathbb{R}^{B \times N \times d}$, where B , N , and d denote the batch size, sequence length, and embedding dimension, respectively. The input is first transposed so that the embedding dimension becomes the convolutional channel dimension. The first 1×1 pointwise convolution then expands the channel dimension to $2d$:

$$\mathbf{Y}_1 = \text{PW}_{\uparrow}(\mathcal{T}(\mathbf{X})) \in \mathbb{R}^{B \times 2d \times N}. \quad (17)$$

The expanded features pass through a 1×5 depthwise convolution, which processes the sequence features independently within each channel to extract information from a short local neighborhood:

$$\mathbf{Y}_2 = \text{DW}_5(\mathbf{Y}_1) \in \mathbb{R}^{B \times 2d \times N}. \quad (18)$$

A ReLU activation is then applied to introduce nonlinearity:

$$\mathbf{Y}_3 = \text{ReLU}(\mathbf{Y}_2). \quad (19)$$

Next, the second 1×1 pointwise convolution integrates features across channels and restores the channel dimension to d :

$$\mathbf{Y}_4 = \text{PW}_{\downarrow}(\mathbf{Y}_3) \in \mathbb{R}^{B \times d \times N}. \quad (20)$$

Finally, the output is transposed back to its original arrangement and combined with the input through a residual connection:

$$\mathbf{X}_S = \text{DSConv-S}(\mathbf{X}) = \mathbf{X} + \mathcal{T}^{-1}(\mathbf{Y}_4). \quad (21)$$

where $\text{PW}_\uparrow$ and $\text{PW}_\downarrow$ denote the pointwise convolutions for channel expansion and restoration, respectively, DW_5 denotes the 1×5 depthwise convolution, and $\mathcal{T}(\cdot)$ and $\mathcal{T}^{-1}(\cdot)$ denote the transpose and inverse transpose operations, respectively.

Computational analysis. With a channel expansion factor of two, the computational cost of DSConv-S mainly comes from two 1×1 pointwise convolutions and one 1×5 depthwise convolution and can be expressed as:

$$C_{\text{in}} \times 2C_{\text{out}} \times D_F \times D_F + 1 \times 5 \times 2C_{\text{out}} \times D_F \times D_F + 2C_{\text{out}} \times C_{\text{out}} \times D_F \times D_F. \quad (22)$$

where $2C_{\text{out}}$ and $D_F \times D_F$ denote the expanded number of channels and the feature map size, respectively.

3.2.3 DSConv-L: Feature refinement over longer time scales

DSConv-S enhances local information before attention, while DSConv-L follows LGFA to further refine the feature representation that combines local information and global context. A 1×1 pointwise convolution first expands the channel dimension by a factor of three to extract rich representations, followed by a 1×31 depthwise convolution that captures local sequential patterns over longer time scales within each channel. A second 1×1 pointwise convolution integrates these features across channels and restores the original channel dimension. The transformation order is the same as in DSConv-S. At the MS-AgentNet Block level, the convolution output is scaled by a learnable factor and added to the original LGFA output through a residual connection, incorporating the further extracted sequence features into the existing representation, as shown in Eq. (12). The small- and large-kernel convolutions thus jointly enrich multi-scale degradation representations for SOH estimation.

Computational analysis. With a channel expansion factor of three, the computational cost of DSConv-L can be expressed as:

$$C_{\text{in}} \times 3C_{\text{out}} \times D_F \times D_F + 1 \times 31 \times 3C_{\text{out}} \times D_F \times D_F + 3C_{\text{out}} \times C_{\text{out}} \times D_F \times D_F. \quad (23)$$

where $3C_{\text{out}}$ and $D_F \times D_F$ denote the expanded number of channels and the feature map size, respectively.

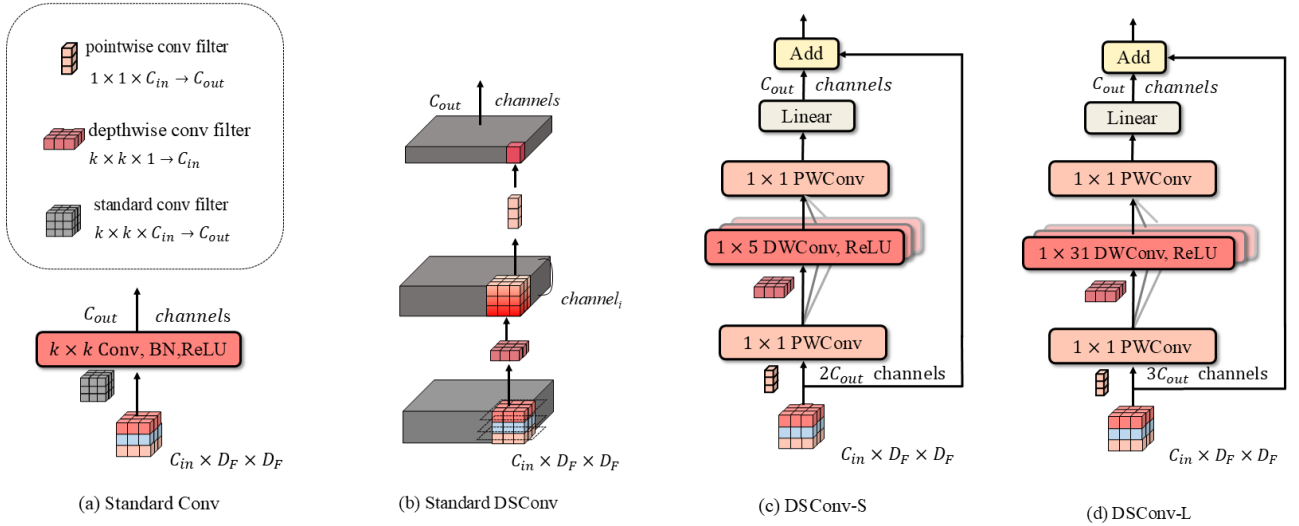


Fig. 6. The fundamental structures of convolutional modules: (a) standard convolution; (b) standard depthwise separable convolution; (c) DSConv-S module; and (d) DSConv-L module.

3.3 The proposed Local-Global Fusion Attention module

This section presents the general form of multi-head self-attention, briefly compares Softmax attention and linear attention, and introduces LGFA, which combines local feature extraction with global interactions that scale linearly with sequence length.

3.3.1 General form of multi-head self-attention

Multi-head self-attention uses multiple parallel attention heads, each calculating correlations between features within a different representation subspace [29]. Given input features $\mathbf{X} \in \mathbb{R}^{N \times d}$, the query, key, and value representations of the i th attention head are:

$$\mathbf{Q}_i = \mathbf{XW}_i^Q, \quad \mathbf{K}_i = \mathbf{XW}_i^K, \quad \mathbf{V}_i = \mathbf{XW}_i^V. \quad (24)$$

where N is the sequence length, d is the input feature dimension, and $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{d \times d_h}$ are learnable linear projection matrices. Here, $i = 1, 2, \dots, h$, h is the number of attention heads, and $d_h = d/h$ is the feature dimension of each head.

When a nonnegative similarity function is used to construct normalized attention weights, the output at the p th position of the i th attention head can be expressed as:

$$\mathbf{O}_{i,p} = \frac{\sum_{j=1}^N \text{Sim}(\mathbf{Q}_{i,p}, \mathbf{K}_{i,j}) \mathbf{V}_{i,j}}{\sum_{j=1}^N \text{Sim}(\mathbf{Q}_{i,p}, \mathbf{K}_{i,j})}, \quad p = 1, 2, \dots, N. \quad (25)$$

where $\mathbf{Q}_{i,p}$ is the query at the p th position in the i th attention head, $\mathbf{K}_{i,j}$ and $\mathbf{V}_{i,j}$ are the key and value at the j th position, respectively, and $\text{Sim}(\cdot, \cdot)$ is a nonnegative similarity function.

The outputs at all positions form the output matrix $\mathbf{O}_i \in \mathbb{R}^{N \times d_h}$ of the i th attention head. The outputs of all heads are then concatenated along the feature dimension. Attention mechanisms mainly differ in their similarity functions and computation order.

3.3.2 Softmax attention and linear attention

Standard Softmax attention calculates correlations between sequence positions through query-key dot products and normalizes the similarity scores using the Softmax function [29]. The output of the i th attention head can be expressed as:

$$\mathbf{O}_i = \text{Softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_h}}\right) \mathbf{V}_i. \quad (26)$$

The exponential mapping and normalization in Softmax assign larger attention weights to more highly correlated query-key pairs. However, computing $\mathbf{Q}_i \mathbf{K}_i^T$ requires pairwise similarity calculations between all queries and keys, forming an $N \times N$ attention score matrix. Thus, the computational complexity of one attention head is $O(N^2 d_h)$, and the memory complexity of the attention score matrix is $O(N^2)$. As the sequence length increases, the computational and storage overhead grows substantially [35].

To address the quadratic computational cost of Softmax attention, linear attention maps queries and keys through a kernel function $\phi(\cdot)$ and rearranges the computation order using the associative property of matrix multiplication [36]. The output of the i th attention head can be written as:

$$\mathbf{O}_i = \frac{\phi(\mathbf{Q}_i) [\phi(\mathbf{K}_i)^T \mathbf{V}_i]}{\phi(\mathbf{Q}_i) [\phi(\mathbf{K}_i)^T \mathbf{1}]}. \quad (27)$$

where $\phi(\cdot)$ is a nonnegative feature mapping, such as $\phi(\mathbf{x}) = \text{ELU}(\mathbf{x}) + 1$, and $\mathbf{1}$ is an all-ones vector used to calculate the corresponding normalization term.

By first calculating $\phi(\mathbf{K}_i)^T \mathbf{V}_i$, linear attention avoids constructing the full $N \times N$ attention matrix. When the

feature mapping dimension equals d_h , its computational complexity is $O(Nd_h^2)$. For a fixed d_h , this complexity is linear in the sequence length N .

Linear attention enables global information interactions at a low computational complexity, but its computation does not explicitly incorporate local neighborhood features [25, 35, 39]. Modeling health indicator sequences constructed from multi-source signals and their derived curves requires both capturing long-term battery degradation trends and retaining fine-grained degradation information in variations between neighboring cycles. Local feature extraction therefore needs to be combined with an efficient attention mechanism to balance degradation representation and computational efficiency.

3.3.3 The proposed LGFA module

To address these requirements, small-kernel depthwise separable convolution (DSConv-S) is combined with ReLU² Agent Attention (RAA) to form the Local-Global Fusion Attention (LGFA) module, as shown in Fig. 7(d). DSConv-S first enhances local information in the health indicator sequences, and the resulting representation is fed into the local and RAA branches. The local branch preserves local degradation information, while the RAA branch establishes relationships across cycles within the input window through agent aggregation and broadcasting. The outputs of the two branches are fused by addition.

(1) ReLU² Agent Attention.

RAA uses a small number of agents to mediate global interactions, avoiding direct pairwise attention between all sequence positions [66]. During agent aggregation, the agents gather information from the keys and values; during information broadcasting, each query retrieves information from the agent context. Together, these two stages enable degradation-related information to be shared among health indicator representations within the input window, thereby establishing cross-cycle context.

In RAA, $\mathbf{X}$ denotes the input features. Within LGFA, this input is the local representation $\mathbf{X}_S$ produced by DSConv-S. To reduce the parameter overhead of query, key, and value generation, RAA uses learnable channel scaling to adjust the relative contributions of different feature channels and splits the scaled features across attention heads:

$$\mathbf{Q}_i = \text{Split}_i(\mathbf{X} \odot \mathbf{s}_q), \quad \mathbf{K}_i = \text{Split}_i(\mathbf{X} \odot \mathbf{s}_k), \quad \mathbf{V}_i = \text{Split}_i(\mathbf{X} \odot \mathbf{s}_v). \quad (28)$$

where $\mathbf{s}_q, \mathbf{s}_k, \mathbf{s}_v \in \mathbb{R}^d$ are learnable channel scaling vectors, $\odot$ denotes element-wise multiplication, and $\text{Split}_i(\cdot)$ divides the feature dimension into h subspaces of dimension d_h and selects the i th subspace.

Compared with standard linear projections for queries, keys, and values, which require approximately $3d^2$ parameters, the three channel scaling vectors contain only $3d$ parameters, reducing the parameter overhead of query, key, and value generation.

RAA uses $n_a = 2$ learnable agents, represented by a matrix $\mathbf{A} \in \mathbb{R}^{n_a \times d}$. Their parameters are learned during training and shared across input samples.

The global agent matrix $\mathbf{A}$ is divided into h subspaces along the feature dimension. The agent representation for the i th attention head is $\mathbf{A}_i = \text{Split}_i(\mathbf{A}) \in \mathbb{R}^{n_a \times d_h}$.

During information broadcasting, each query assigns broadcasting weights according to how it matches the two agents. Standard Agent Attention uses Softmax normalization in both aggregation and broadcasting. When all scores in a row are reduced by a common positive scaling factor, the relative ratios between positive scores remain unchanged, but the weights approach a uniform distribution. The corresponding query therefore reads an increasingly equal mixture of the two agent contexts. To preserve the distinction between positive matches under such scale changes, RAA uses ReLU² row normalization in both stages:

$$[\mathcal{R}_2(\mathbf{Z})]_{r,c} = \begin{cases} \frac{\text{ReLU}(\mathbf{Z}_{r,c})^2}{\sum_{j=1}^M \text{ReLU}(\mathbf{Z}_{r,j})^2}, & \sum_{j=1}^M \text{ReLU}(\mathbf{Z}_{r,j})^2 > 0, \\ \frac{1}{M}, & \sum_{j=1}^M \text{ReLU}(\mathbf{Z}_{r,j})^2 = 0. \end{cases} \quad (29)$$

where M is the number of elements normalized in each row: $M = N$ during agent aggregation and $M = n_a$ during

information broadcasting. ReLU sets nonpositive similarity scores to zero. For a row containing positive scores, the squared common positive scaling factor cancels between the numerator and denominator. Thus, the weight distribution remains unchanged when all scores are scaled by the same positive factor. Squaring further increases the weight ratio between larger and smaller positive scores, giving greater emphasis to higher positive similarity scores. When a row contains no positive similarity scores, $\mathcal{R}_2(\cdot)$ returns a uniform distribution to keep the normalization valid.

During agent aggregation, $\mathbf{A}_i$ acts as the query to gather context from the sequence keys and values:

$$\Phi_{k,i} = \mathcal{R}_2 \left(\frac{\mathbf{A}_i \mathbf{K}_i^T}{\sqrt{d_h}} \right), \quad \mathbf{V}_{A,i} = \Phi_{k,i} \mathbf{V}_i. \quad (30)$$

where $\Phi_{k,i} \in \mathbb{R}^{n_a \times N}$ contains the agents' aggregation weights for each sequence position, and $\mathbf{V}_{A,i} \in \mathbb{R}^{n_a \times d_h}$ is the resulting agent context.

During information broadcasting, each query position reads information from the agent context according to its own features:

$$\Phi_{q,i} = \mathcal{R}_2 \left(\frac{\mathbf{Q}_i \mathbf{A}_i^T}{\sqrt{d_h}} \right), \quad \mathbf{O}_i = \Phi_{q,i} \mathbf{V}_{A,i}. \quad (31)$$

where $\Phi_{q,i} \in \mathbb{R}^{N \times n_a}$ contains the query-specific broadcasting weights over the agents, and $\mathbf{O}_i \in \mathbb{R}^{N \times d_h}$ is the output of the i th attention head. ReLU² row normalization converts query-agent similarity scores into broadcasting weights, which combine the two agent contexts to form the corresponding global context representation.

Equations (30) and (31) show that the agent-mediated sequence interaction in the i th attention head can be represented by the equivalent mapping $\Phi_{q,i} \Phi_{k,i} \in \mathbb{R}^{N \times N}$, whose rank is at most the number of agents n_a . This equivalent matrix is not explicitly constructed; the computation proceeds through agent aggregation and information broadcasting.

The outputs of all attention heads are concatenated along the feature dimension. A learnable output scaling vector $\mathbf{s}_o$ then adjusts the channel responses, and the result is added to the input features through a residual connection:

$$\text{RAA}(\mathbf{X}) = \mathbf{X} + \text{Concat}(\mathbf{O}_1, \dots, \mathbf{O}_h) \odot \mathbf{s}_o. \quad (32)$$

where $\text{Concat}(\cdot)$ concatenates the outputs of all attention heads along the feature dimension, and $\mathbf{s}_o \in \mathbb{R}^d$ is a learnable output scaling vector.

RAA generates attention weight matrices of size $n_a \times N$ and $N \times n_a$ during aggregation and broadcasting, respectively. The computational complexity is therefore $O(Nn_ad_h)$ for one attention head and $O(Nn_ad)$ for all h heads, while the memory required for the attention weights is $O(hNn_a)$. For fixed d , h , and n_a , the computational cost of the attention interactions and the memory required for the attention weights both grow linearly with the sequence length N .

(2) Local-global feature fusion.

The input features $\mathbf{X}$ first pass through DSConv-S to obtain the local representation $\mathbf{X}_S = \text{DSConv-S}(\mathbf{X})$, which is then fed into the local and RAA branches. The local branch applies layer normalization to $\mathbf{X}_S$, while the RAA branch establishes cross-position feature interactions through agent aggregation and information broadcasting. The fused output $\mathbf{X}_F$ of LGFA is calculated as:

$$\mathbf{X}_F = \text{LN}(\mathbf{X}_S) + W_a \text{Dropout}(\text{RAA}(\mathbf{X}_S)). \quad (33)$$

where $\mathbf{X}_F$ is the fused LGFA output, $\text{LN}(\mathbf{X}_S)$ is the normalized local branch representation, and $\text{RAA}(\mathbf{X}_S)$ is the RAA branch output. $\text{Dropout}(\cdot)$ provides regularization during training [67], and W_a is a learnable scaling factor that adjusts the contribution of the RAA branch.

This additive fusion combines the local degradation features extracted by DSConv-S with the global context established by RAA, providing a joint local-global representation for SOH estimation while maintaining linear computational complexity with respect to the sequence length N .

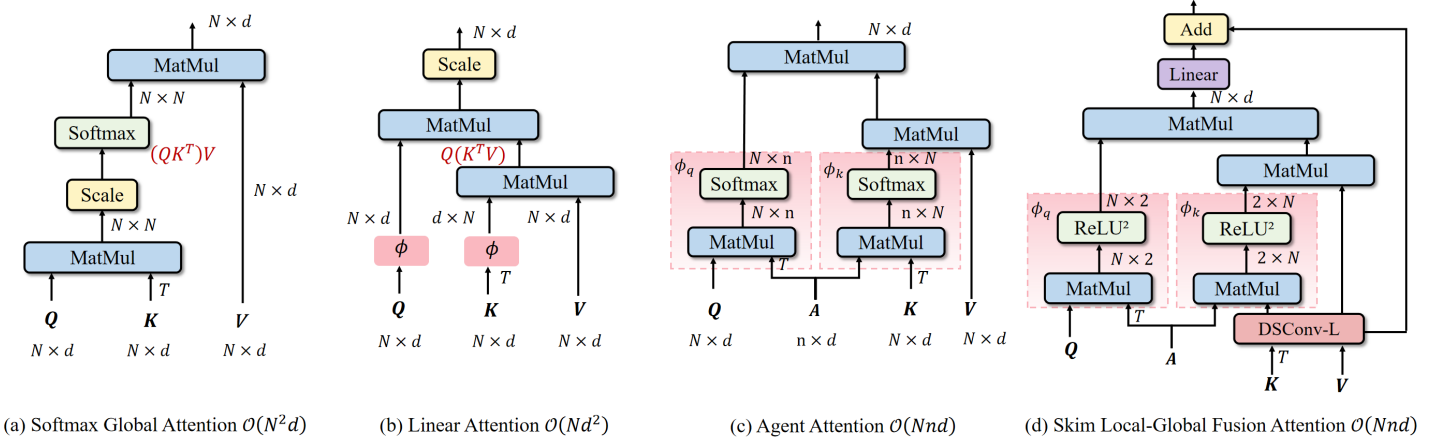


Fig. 7. Comparison of (a) Softmax attention, (b) linear attention, (c) Agent Attention, and (d) the proposed LGFA module.

4. Experimental results and analysis

This section systematically evaluates MS-AgentNet on four public battery datasets. The experiments cover HI effectiveness, model performance comparisons, module ablation and complexity analysis, and cross-domain adaptation. The HI experiments examine whether the selected features can represent battery degradation patterns and thus provide suitable model inputs. The model comparisons assess estimation accuracy and stability across cells. Ablation studies further examine the individual and complementary effects of multi-scale local feature extraction and cross-cycle context modeling, and compare different convolutional configurations to assess the complementary roles of DSConv-S and DSConv-L; complexity analysis evaluates computational and storage overhead. Finally, cross-dataset transfer experiments examine the model’s transfer performance across different data domains. Together, these experiments provide a comprehensive evaluation of the effectiveness and applicability of MS-AgentNet for battery SOH estimation.

Model training is conducted on a compute instance with 25 vCPUs provided by an Intel Xeon Platinum 8470Q processor, one NVIDIA GeForce RTX 5090 GPU with 32 GB of graphics memory, and 90 GB of RAM. The instance runs Ubuntu 22.04. The prediction models are implemented using PyTorch 2.12.1 and Python 3.12, with CUDA 13.0.

4.1 Evaluation metrics

Four commonly used metrics are selected to evaluate SOH estimation performance: mean absolute error (MAE), mean absolute percentage error (MAPE), root mean square error (RMSE), and the coefficient of determination (R^2) [68].

MAE is the mean absolute difference between predicted and true SOH over all evaluation samples and reflects the overall error level. It does not assign extra weight to individual large errors and provides a direct measure of overall estimation accuracy.

MAPE normalizes absolute errors by the true SOH, allowing estimation performance to be compared across datasets. When the true SOH approaches zero, MAPE may become much larger and should be considered together with MAE and RMSE.

RMSE is the square root of the mean squared difference between predicted and true SOH. Compared with MAE, it assigns greater weight to large prediction deviations and thus further reflects the model’s ability to limit large errors.

R^2 measures how well the model fits variations in the true SOH. A value closer to 1 indicates a better fit to the overall degradation trend.

The four metrics are defined as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (34)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (35)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (36)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (37)$$

where y_i and $\hat{y}_i$ are the true and predicted SOH of the i th sample, respectively, $\bar{y}$ is the mean true SOH of the evaluation samples, and n is the total number of evaluation samples.

4.2 HI selection results and effectiveness analysis

To evaluate how effectively group-level HI selection identifies inputs that remain stable across cells, this section compares SOH estimation performance with different HIs on the Oxford dataset. Based on the PCC and SCC of candidate HIs on Cell1, the indicator with the highest combined correlation score is selected from each of the IC, DTV, and DTC categories, giving HI4, HI9, and HI11, respectively. All three combined correlation scores exceed 0.94. These HIs and the CCCT indicator HI1, which is obtained through group-level selection, are separately fed into MS-AgentNet. The input combining HI1, HI4, HI9, and HI11 is denoted as Fusion. The validation result on Cell2 and the test results on Cell3–Cell8 with different inputs are given in Table 6.

The results show that using HI1 consistently yields high estimation accuracy across Cell2–Cell8, with the lowest average MAE, RMSE, and MAPE among the five input schemes. Its average MAPE is 0.00615, or 0.615%. When all four HIs are used together, Fusion ranks fourth for all three average errors, showing that directly combining multiple types of HIs does not provide further performance gains.

Moreover, the representational ability of the same HI differs across cells. For example, HI9 yields lower estimation errors than HI11 on Cell7, whereas the opposite is observed on Cell8. In contrast, HI1 obtained through group-level selection achieves the lowest MAE, RMSE, and MAPE on all seven cells. These results show that a high correlation on one cell does not guarantee the same representational ability on other cells in the group. The group-selected HI1 provides a more stable input for SOH estimation on this group of Oxford cells.

Table 6

SOH estimation errors of different health-indicator inputs on the Oxford dataset.

Cell	MAE					RMSE					MAPE				
	CCCT	IC	DTV	DTC	Fusion	CCCT	IC	DTV	DTC	Fusion	CCCT	IC	DTV	DTC	Fusion
Cell2*	0.00437	0.01607	0.01941	0.01458	0.01256	0.00706	0.02238	0.02145	0.01890	0.01634	0.00535	0.02036	0.02344	0.01810	0.01542
Cell3	0.00295	0.00456	0.08852	0.01490	0.03835	0.00366	0.00576	0.09371	0.01770	0.04087	0.00351	0.00527	0.10174	0.01765	0.04466
Cell4	0.00838	0.00934	0.01614	0.02349	0.01751	0.00898	0.01249	0.01679	0.02505	0.01898	0.00988	0.01115	0.01867	0.02705	0.02027
Cell5	0.00314	0.00819	0.03244	0.02131	0.02085	0.00362	0.00903	0.03532	0.02267	0.02353	0.00357	0.00924	0.03672	0.02428	0.02378
Cell6	0.00619	0.00941	0.03163	0.00875	0.01381	0.00860	0.01302	0.03414	0.01276	0.01670	0.00733	0.01087	0.03556	0.01030	0.01604
Cell7	0.00777	0.01408	0.02278	0.02534	0.01837	0.00804	0.01499	0.02650	0.03389	0.02344	0.00894	0.01594	0.02619	0.03028	0.02154
Cell8	0.00389	0.00967	0.05388	0.02084	0.02694	0.00469	0.01099	0.05597	0.02659	0.03201	0.00445	0.01102	0.06297	0.02530	0.03159
Average	0.00524	0.01019	0.03783	0.01846	0.02120	0.00638	0.01267	0.04055	0.02251	0.02455	0.00615	0.01198	0.04361	0.02185	0.02476

Note: Cell2* is the validation cell for hyperparameter selection. CCCT, IC, DTV, and DTC correspond to HI1, HI4, HI9, and HI11, respectively.

4.3 Training robustness and hyperparameter analysis

To further examine the training stability and parameter settings of the lightweight MS-AgentNet model, this study analyzes agent matrix initialization, model convergence, and hyperparameter configuration.

4.3.1 Agent matrix initialization and convergence analysis

RAA represents the agents as a static learnable matrix $\mathbf{A} \in \mathbb{R}^{n_a \times d}$ that is independent of the input and used for global feature aggregation. The number of agents is fixed at $n_a = 2$. The matrix parameters are initialized from a normal distribution with a mean of 0 and a standard deviation of 0.02:

$$A_{ij} \sim \mathcal{N}(0, 0.02^2) \quad (38)$$

where A_{ij} is the parameter of the i th agent in the j th feature dimension. This initialization gives different agents small, nonidentical initial parameters, avoiding identical initial states.

Table 7 compares the three initialization schemes. Their R^2 values range from 0.98527 to 0.98645, RMSE from 0.00774 to 0.00805, and MAE from 0.00485 to 0.00499, indicating similar prediction performance. Truncated normal initialization controls parameter magnitudes by restricting the sampling range, while Xavier uniform initialization adjusts parameter variance according to the input and output dimensions of a layer. In comparison, ordinary normal initialization directly provides small, zero-centered random parameters for the static agent matrix without additional range restrictions or dimension-based scaling. It is therefore used as the default setting.

Model convergence is evaluated through both convergence speed and loss fluctuations in the late training stage. The convergence threshold epoch is defined as the first epoch at which the training loss falls below 10% of the first-epoch loss. The mean and standard deviation of the loss over the final 20 epochs describe the late-stage convergence state. As shown in Table 8, every training run reaches the predefined threshold. The median convergence threshold epochs are 26 and 7 on the Oxford and MIT datasets, respectively, and the median standard deviations of the late-stage loss are 2.38×10^{-4} and 4.84×10^{-6} . These results show that MS-AgentNet converges stably under the current experimental settings, with small loss fluctuations in the late training stage.

Table 7

Initialization strategy comparison on the Oxford validation cell (Cell2).

Initialization	R^2	RMSE	MAE
	mean $\pm$ SD	mean $\pm$ SD	mean $\pm$ SD
Normal	0.98606	0.00784	0.00490
$\mathcal{N}(0, 0.02^2)$	± 0.00411	± 0.00116	± 0.00054
Truncated normal	0.98645	0.00774	0.00485
$(\pm 2\sigma)$	± 0.00371	± 0.00103	± 0.00048
Xavier uniform	0.98527	0.00805	0.00499
	± 0.00462	± 0.00126	± 0.00058

Table 8

Convergence characteristics of MS-AgentNet on the Oxford and MIT datasets.

Dataset	10% threshold epoch median [range]	Late loss mean median	Late loss SD median
Oxford	26 [18–35]	8.14×10^{-4}	2.38×10^{-4}
MIT	7 [6–8]	5.84×10^{-5}	4.84×10^{-6}

4.3.2 Hyperparameter configuration

The main hyperparameters of MS-AgentNet are the learning rate, network depth L , and embedding dimension d . The hyperparameter search space is listed in Table 10. For each dataset, one cell in the feature-development set is used to learn model parameters, while a second cell is used for validation and comparison to select the best-performing hyperparameter configuration. The training and validation cells are Cell1 and Cell2 for Oxford, CS2_36 and CS2_37 for CALCE CS2, CX2_36 and CX2_37 for CALCE CX2, and b3c8 and b3c13 for MIT/Severson, respectively.

The layer configurations of the five models are given in Table 9.

Table 9

Layer configurations of different deep learning models.

MS-AgentNet	CNN-Transformer	Transformer	CNN-LSTM	LSTM
Input	Input	Input	Input	Input
Embedding layer (16)	Embedding layer (16)	Embedding layer (16)	Conv1D + ReLU	Linear layer (16)
LGFA module	Positional encoder	Positional encoder	MaxPooling1D	5× LSTM layer (16)
DConv-L module	Softmax attention	Softmax attention	3× Conv1D + ReLU	Linear layer
Add & Norm	2× Conv1D + ReLU	Add & Norm	3× LSTM layer (16)	
FFN	MLP	MLP	Linear layer	
Add & Norm	Transformer decoder	Add & Norm		
Linear readout layer	Linear layer	Transformer decoder		
		Linear layer		

MLP denotes multilayer perceptron.

Once selected, the model configurations remain fixed. In the subsequent SOH estimation comparisons, MS-AgentNet uses a learning rate of 0.001, network depth $L = 1$, and embedding dimension $d = 16$.

Table 10

Hyperparameter search space of MS-AgentNet.

Hyperparameter	Search space/Values
Learning rate	0.001, 0.01
Network depth L	1, 2, 4, 8
Embedding dimension d	16, 32, 64, 128

4.4 Comparison of SOH estimation accuracy and cross-cell generalization

Based on the selection results in Section 2.3, the inputs are HI1 for Oxford, HI1 and HI2 for CALCE CS2, HI13 for CALCE CX2, and HI14 and HI15 for MIT/Severson. With these HI inputs, the SOH estimation performance of MS-AgentNet is further compared with that of four baseline models.

4.4.1 Estimation accuracy on the Oxford dataset

Fig. 8 and Table 11 present the validation results on Oxford Cell2 and the test results on Cell3–Cell8 for MS-AgentNet, CNN-Transformer, CNN-LSTM, Transformer, and LSTM.

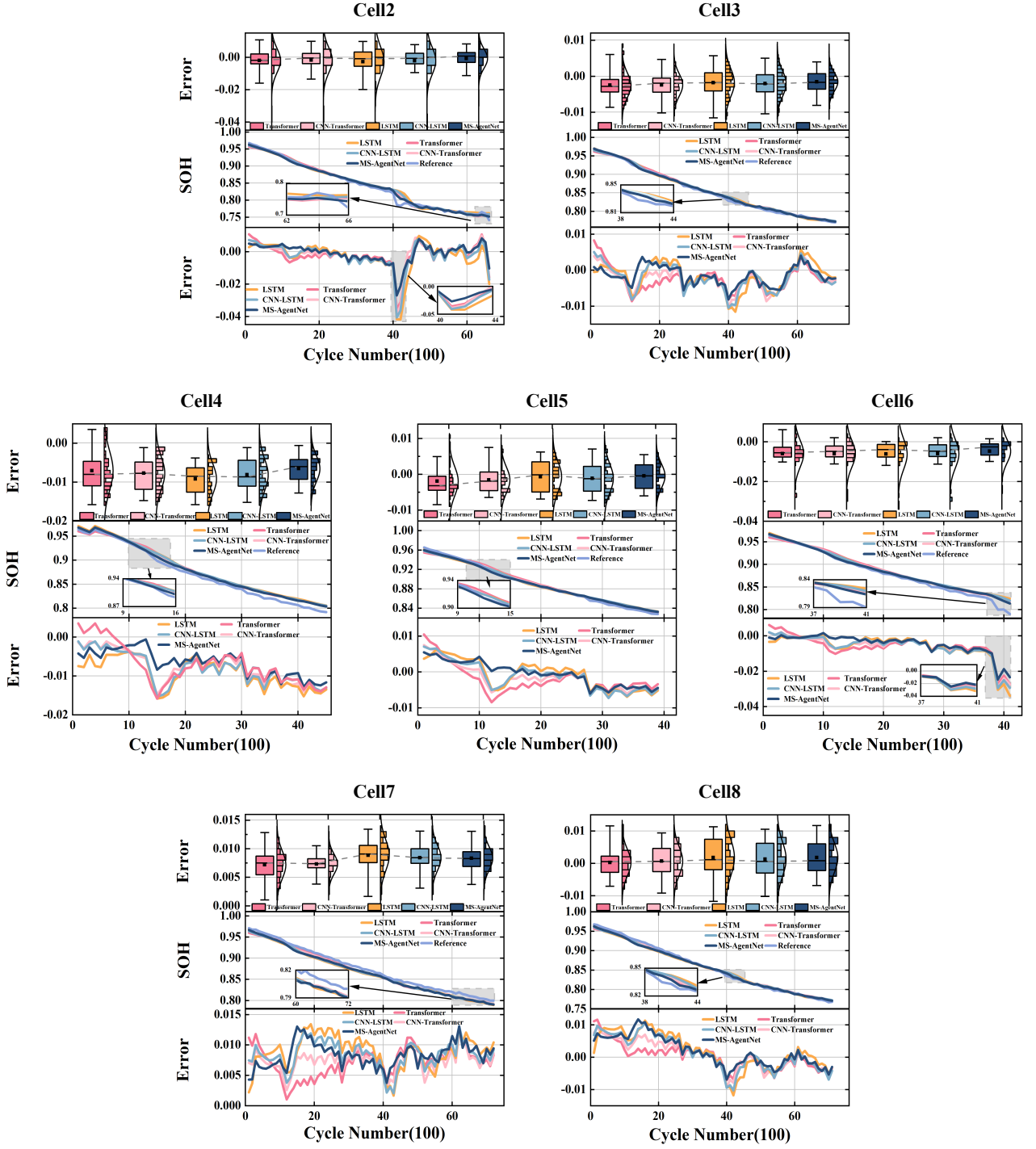


Fig. 8. SOH estimation results of different models on the Oxford dataset.

Overall, all models track the capacity fade trends, but their predictions differ in regions with local variations. In particular, the local deviations in Cell4 and the rapid decline in the later stage of Cell6 lead to different degrees of tracking error. The enlarged views show that MS-AgentNet still tracks the true SOH well in these regions. Its predictions also closely follow the true SOH of Cell3 and Cell5, whose degradation trajectories are relatively smooth.

The quantitative results show that MS-AgentNet achieves the best R^2 , MAE, MAPE, and RMSE on Cell2, Cell3, Cell5, and Cell6, as well as the highest R^2 and lowest RMSE on Cell4. Averaged over Cell2–Cell8, its R^2 is 0.98379, and its MAE, MAPE, and RMSE are 0.00524, 0.00615, and 0.00638, respectively, ranking first among the five models for all four metrics. Compared with CNN-Transformer, CNN-LSTM, Transformer, and LSTM, its average MAE is reduced by 2.42%, 8.87%, 4.20%, and 12.52%, its average MAPE by 2.38%, 8.48%, 3.76%, and 12.52%, and its average RMSE by 5.06%, 11.39%, 5.62%, and 15.83%, respectively. These results show that MS-AgentNet achieves

high SOH estimation accuracy and good prediction robustness on the Oxford dataset.

Table 11

SOH estimation errors of different models on the Oxford dataset.

Cell	Method	R^2	MAE	MAPE	RMSE
Cell2*	MS-AgentNet	0.98890	0.00437	0.00535	0.00706
	CNN-Transformer	0.98532	0.00497	0.00609	0.00812
	CNN-LSTM	0.98341	0.00521	0.00635	0.00863
	Transformer	0.98571	0.00512	0.00622	0.00800
	LSTM	0.97880	0.00590	0.00725	0.00977
Cell3	MS-AgentNet	0.99638	0.00295	0.00351	0.00366
	CNN-Transformer	0.99523	0.00335	0.00396	0.00420
	CNN-LSTM	0.99501	0.00343	0.00404	0.00429
	Transformer	0.99472	0.00362	0.00423	0.00440
	LSTM	0.99529	0.00327	0.00390	0.00418
Cell4	MS-AgentNet	0.97052	0.00838	0.00988	0.00898
	CNN-Transformer	0.96942	0.00819	0.00969	0.00914
	CNN-LSTM	0.96712	0.00871	0.01026	0.00949
	Transformer	0.96931	0.00820	0.00966	0.00916
	LSTM	0.96267	0.00937	0.01103	0.01011
Cell5	MS-AgentNet	0.99257	0.00314	0.00357	0.00362
	CNN-Transformer	0.99002	0.00344	0.00389	0.00417
	CNN-LSTM	0.98912	0.00351	0.00397	0.00434
	Transformer	0.98727	0.00391	0.00437	0.00468
	LSTM	0.99036	0.00355	0.00400	0.00412
Cell6	MS-AgentNet	0.96832	0.00619	0.00733	0.00860
	CNN-Transformer	0.96442	0.00646	0.00764	0.00911
	CNN-LSTM	0.96226	0.00650	0.00771	0.00939
	Transformer	0.96402	0.00696	0.00815	0.00917
	LSTM	0.95868	0.00626	0.00747	0.00983
Cell7	MS-AgentNet	0.97566	0.00777	0.00894	0.00804
	CNN-Transformer	0.97781	0.00731	0.00841	0.00766
	CNN-LSTM	0.97081	0.00841	0.00964	0.00880
	Transformer	0.97750	0.00719	0.00830	0.00769
	LSTM	0.96768	0.00886	0.01016	0.00926
Cell8	MS-AgentNet	0.99419	0.00389	0.00445	0.00469
	CNN-Transformer	0.99434	0.00384	0.00442	0.00462
	CNN-LSTM	0.99207	0.00445	0.00508	0.00548
	Transformer	0.99528	0.00326	0.00377	0.00422
	LSTM	0.99115	0.00474	0.00539	0.00579
Average	MS-AgentNet	0.98379	0.00524	0.00615	0.00638
	CNN-Transformer	0.98237	0.00537	0.00630	0.00672
	CNN-LSTM	0.97997	0.00575	0.00672	0.00720
	Transformer	0.98197	0.00547	0.00639	0.00676
	LSTM	0.97780	0.00599	0.00703	0.00758

Note: Cell2* denotes the validation cell used for hyperparameter selection, and its values are averages of five independent runs. The best results are shown in bold.

4.4.2 Cross-cell generalization on the CALCE and MIT datasets

This section further compares MS-AgentNet with the four baseline models on the CALCE CS2, CALCE CX2, and MIT/Severson datasets. CS2_37, CX2_37, and b3c13 are the validation cells, while CS2_38, CX2_38, and b3c29 are the test cells. These datasets differ from Oxford in battery chemistries and charge-discharge protocols and show diverse degradation dynamics, including smooth capacity fade, transitions between stages, and accelerated late-life degradation. The SOH estimation results are shown in Fig. 9.

Overall, all models track the capacity fade trends of different cells, but their prediction performance differs at local nonlinear transitions and during accelerated late-life degradation. In particular, CX2_38 undergoes several transitions

between stages and shows a clear nonlinear degradation tail. The enlarged views show that prediction deviations gradually increase during accelerated degradation, while MS-AgentNet still tracks the accelerated degradation trajectory well. For CS2_38 and b3c29, whose degradation trajectories are more continuous, MS-AgentNet also maintains close agreement with the true SOH.

Table 12 lists the SOH estimation errors of all models on the three validation cells and three test cells. MS-AgentNet achieves the best R^2 , MAE, MAPE, and RMSE on five of them. Notably, on the test cell CX2_38, which shows transitions between stages and a nonlinear degradation tail, its MAPE is only 0.037673, a reduction of 59.75% and 51.80% compared with CNN-Transformer and Transformer, respectively. This shows that the model maintains low prediction errors during accelerated late-life degradation.

Across the six cells, MS-AgentNet achieves an average R^2 of 0.9896 and average MAE, MAPE, and RMSE of 0.00707, 0.013335, and 0.01188, respectively, ranking first among the five models for all four metrics. Compared with CNN-Transformer, CNN-LSTM, Transformer, and LSTM, its average MAE is reduced by 13.50%, 26.02%, 9.11%, and 22.94%, its average MAPE by 42.19%, 60.08%, 34.49%, and 58.17%, and its average RMSE by 12.04%, 26.52%, 7.97%, and 23.09%, respectively. On the three test cells, MS-AgentNet maintains low estimation errors during smooth capacity fade, transitions between stages, and accelerated late-life degradation, showing good prediction robustness and cross-cell generalization capability.

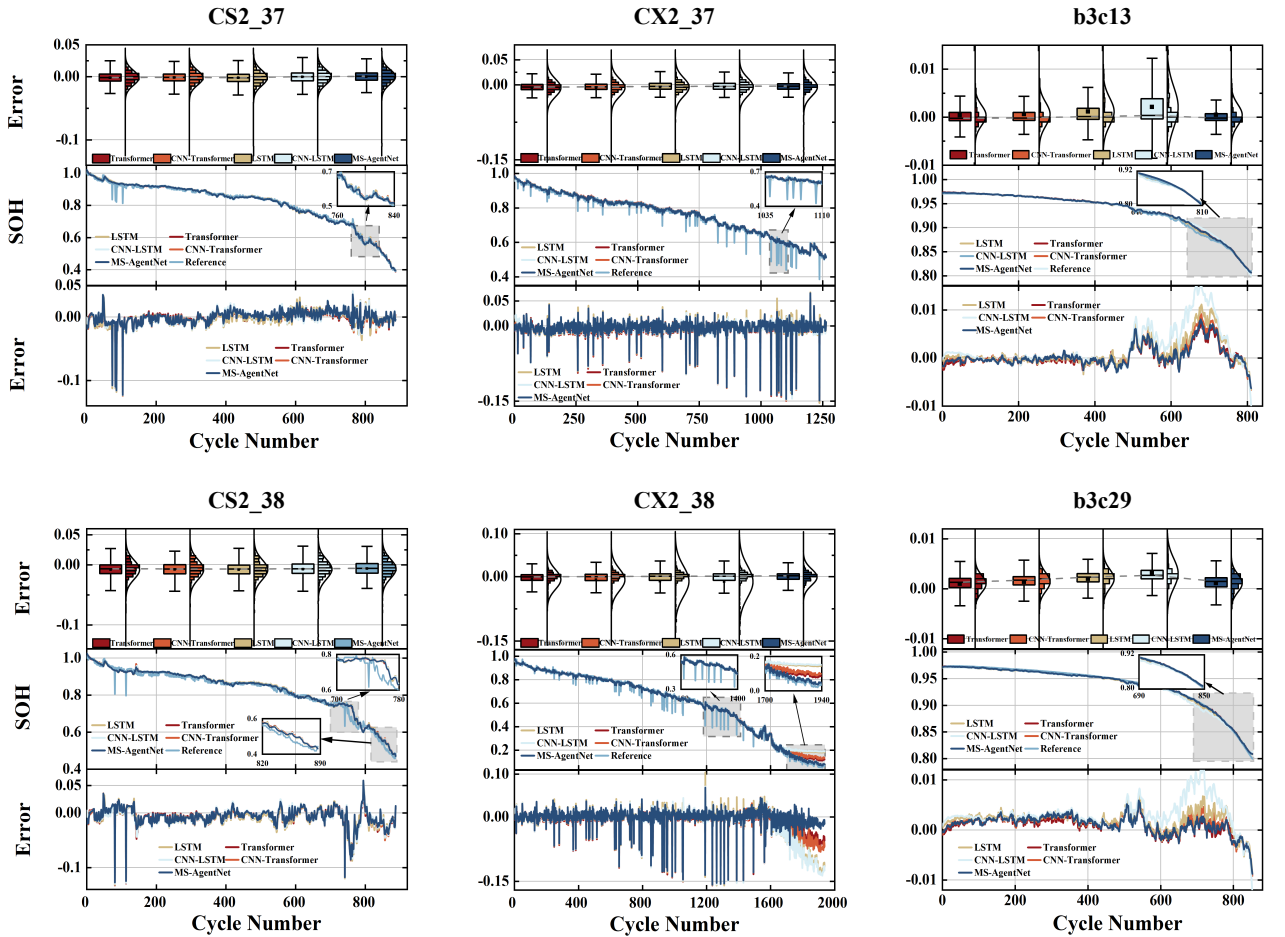


Fig. 9. SOH estimation results of different models on the CALCE and MIT/Severson datasets.

Table 12

SOH estimation errors of different models on the CALCE and MIT/Severson datasets.

Dataset	Method	R^2	MAE	MAPE	RMSE
CALCE CS2 (CS2_37*)	MS-AgentNet	0.9925	0.00733	0.009443	0.01172
	CNN-Transformer	0.9923	0.00742	0.009532	0.01182
	CNN-LSTM	0.9917	0.00800	0.010232	0.01229
	Transformer	0.9923	0.00737	0.009449	0.01182
	LSTM	0.9915	0.00795	0.010106	0.01246
CALCE CS2 (CS2_38)	MS-AgentNet	0.9798	0.01201	0.015984	0.01741
	CNN-Transformer	0.9779	0.01261	0.016928	0.01823
	CNN-LSTM	0.9778	0.01249	0.016680	0.01826
	Transformer	0.9787	0.01216	0.016434	0.01789
	LSTM	0.9776	0.01280	0.017085	0.01836
CALCE CX2 (CX2_37*)	MS-AgentNet	0.9765	0.00916	0.013328	0.01845
	CNN-Transformer	0.9753	0.00981	0.014209	0.01891
	CNN-LSTM	0.9759	0.00974	0.014147	0.01868
	Transformer	0.9748	0.01004	0.014510	0.01909
	LSTM	0.9752	0.00987	0.014322	0.01895
CALCE CX2 (CX2_38)	MS-AgentNet	0.9947	0.01059	0.037673	0.01928
	CNN-Transformer	0.9895	0.01536	0.093595	0.02693
	CNN-LSTM	0.9785	0.02072	0.152435	0.03866
	Transformer	0.9916	0.01378	0.078160	0.02428
	LSTM	0.9804	0.02008	0.145055	0.03704
MIT/Severson (b3c13*)	MS-AgentNet	0.9967	0.00148	0.001622	0.00229
	CNN-Transformer	0.9954	0.00167	0.001832	0.00262
	CNN-LSTM	0.9855	0.00284	0.003121	0.00476
	Transformer	0.9968	0.00146	0.001599	0.00220
	LSTM	0.9940	0.00184	0.002020	0.00309
MIT/Severson (b3c29)	MS-AgentNet	0.9972	0.00183	0.001960	0.00215
	CNN-Transformer	0.9960	0.00214	0.002302	0.00254
	CNN-LSTM	0.9879	0.00353	0.003808	0.00438
	Transformer	0.9971	0.00184	0.001976	0.00219
	LSTM	0.9952	0.00249	0.002667	0.00280
Average	MS-AgentNet	0.9896	0.00707	0.013335	0.01188
	CNN-Transformer	0.9877	0.00817	0.023066	0.01351
	CNN-LSTM	0.9829	0.00955	0.033404	0.01617
	Transformer	0.9885	0.00778	0.020355	0.01291
	LSTM	0.9856	0.00917	0.031876	0.01545

Note: CS2_37*, CX2_37*, and b3c13* denote the validation cells used for hyperparameter selection. The best results are shown in bold.

4.5 Cross-dataset transfer experiments

When the source and target domains come from different datasets, differences in chemistry, operating protocols, and degradation trajectories may cause domain shift [57, 69]. To examine the cross-dataset transfer performance of MS-AgentNet, this study conducts source-only evaluation and few-shot adaptation experiments. In source-only evaluation, the model is trained only on the source-domain reference cell and directly tested on the target-domain cell without updating its parameters using target-domain data. In few-shot adaptation, the feature extraction modules are frozen, and the first 30% of cycle data from the target-domain reference cell are used to update the readout layer before evaluation

on the target-domain test cell [57, 70].

CALCE CS2, CALCE CX2, and Oxford are treated as three data domains. CS2_36, CX2_36, and Oxford Cell1 are used for source-domain training or target-domain adaptation, while CS2_38, CX2_38, and Oxford Cell3 are used only for target-domain testing. CALCE CS2 and CALCE CX2 both use LCO chemistry, whereas Oxford uses a blended NCO-LCO chemistry. The experiments therefore cover both same-chemistry and cross-chemistry transfer. To keep the input definitions consistent between the source and target domains, all transfer directions use two HIs: $CCCT(3.8, 4.0)$ and $Q_{dch}(3.8, 3.4)/C_{rated}$.

4.5.1 Performance across transfer directions

As shown in Table 13, CS2→CX2 and CX2→CS2 achieve R^2 values of 0.8421 and 0.7847 and MAE values of 0.0697 and 0.0388, respectively, indicating that the model still represents degradation trends well when transferred between the two CALCE domains. In contrast, the four transfer directions involving Oxford yield MAE values of 0.1530–0.6346 and R^2 values below 0, showing that direct transfer performance varies considerably with the domain pair and transfer direction.

The few-shot adaptation results are presented in Table 14. After adaptation, MAE, MAPE, and RMSE decrease in all six transfer directions. For MAE, the reductions are 21.91%–45.91% between the two CALCE domains, 86.84%–87.61% for CALCE→Oxford, and 13.27%–31.21% for Oxford→CALCE. After adaptation, R^2 reaches 0.9566 for CS2→CX2 and 0.8806 for CX2→CS2 but remains below 0 in all four directions involving Oxford. This indicates that cross-dataset transfer involving Oxford is more challenging than transfer between the two CALCE domains.

Table 13

Results of source-only evaluation across datasets.

Source	Target	R^2	MAE	MAPE	RMSE
CS2	CX2	0.8421	0.0697	0.3974	0.1051
CS2	Oxford	−60.5701	0.4761	0.5573	0.4778
CX2	CS2	0.7847	0.0388	0.0557	0.0568
CX2	Oxford	−108.6169	0.6346	0.7423	0.6375
Oxford	CS2	−1.5309	0.1530	0.2196	0.1950
Oxford	CX2	−1.4786	0.3528	1.5040	0.4164

Table 14

Transfer results after 30% target-domain adaptation.

Source	Target	R^2	MAE	MAPE	RMSE
CS2	CX2	0.9566	0.0377	0.1830	0.0533
CS2	Oxford	−0.2679	0.0590	0.0722	0.0685
CX2	CS2	0.8806	0.0303	0.0406	0.0422
CX2	Oxford	−1.6085	0.0835	0.1024	0.0982
Oxford	CS2	−1.1030	0.1327	0.1935	0.1777
Oxford	CX2	−0.4809	0.2427	1.2100	0.3218

4.5.2 Effect of the target-domain adaptation ratio

With CS2_36 and CX2_36 used as the respective source domains, the readout layer is adapted with early-cycle data from Oxford Cell1 at adaptation ratios of 10%, 30%, 50%, and 70%. The results are given in Table 15.

As the adaptation ratio increases from 10% to 70%, MAE, MAPE, and RMSE decrease continuously for both source-domain settings. With CS2 as the source domain, MAE decreases from 0.0831 to 0.0252, while R^2 increases from −1.3934 to 0.7650 and becomes positive at an adaptation ratio of 50%. With CX2 as the source domain, MAE decreases from 0.1095 to 0.0559, while R^2 increases from −3.2217 to −0.1268 at an adaptation ratio of 70%. At all four adaptation ratios, the CS2 source domain yields higher R^2 values than the CX2 source domain.

A comparison of adjacent adaptation ratios shows that increasing the ratio from 30% to 50% reduces MAE by 0.0224 and 0.0221 for the CS2 and CX2 source domains, respectively. Increasing it from 50% to 70% gives smaller

reductions of 0.0114 and 0.0055. These results show that adding target-domain observations continuously improves adaptation performance on Oxford. At the same adaptation ratio, the CS2 source domain consistently yields lower estimation errors and higher R^2 , further showing the effect of source-domain selection on cross-dataset adaptation performance.

Table 15

Transfer results with different amounts of target-domain data.

Transfer	Target data (%)	R^2	MAE	MAPE	RMSE
CS2 → Oxford	10	−1.3934	0.0831	0.1015	0.0942
CS2 → Oxford	30	−0.2679	0.0590	0.0722	0.0685
CS2 → Oxford	50	0.5094	0.0366	0.0445	0.0426
CS2 → Oxford	70	0.7650	0.0252	0.0305	0.0293
CX2 → Oxford	10	−3.2217	0.1095	0.1339	0.1251
CX2 → Oxford	30	−1.6085	0.0835	0.1024	0.0982
CX2 → Oxford	50	−0.3836	0.0614	0.0747	0.0716
CX2 → Oxford	70	−0.1268	0.0559	0.0677	0.0646

4.6 Module ablation and model complexity analysis

The preceding SOH estimation results show that the batteries exhibit different degradation characteristics, including smooth capacity fade, local variations, stage transitions, and accelerated late-life degradation, and that the full MS-AgentNet generally maintains low estimation errors across these scenarios. However, comparisons of complete models do not distinguish the specific roles of different structures in degradation feature modeling. Module ablation and complexity analyses are therefore conducted. The ablation studies remove and combine multi-scale DSConv and RAA to examine the individual and complementary effects of local degradation information extraction and cross-cycle context modeling, and further analyze the complementary roles of DSConv-S and DSConv-L at different feature-processing stages; complexity analysis quantifies the computational and storage overhead of the resulting architecture.

4.6.1 Module ablation analysis

Efficient attention reduces the computational overhead of standard attention by compressing information interactions [35, 36, 71, 72], but this process may weaken the representation of fine-grained local degradation information [25]. Directly introducing standard convolutions can enhance local feature modeling but increases the parameter count and computational load [65]. To balance local degradation representation and computational efficiency, MS-AgentNet uses multi-scale DSConv to extract fine-grained local degradation information from health indicator sequences and RAA to establish cross-cycle context within the input window. Four variants, M1–M4, are used to examine their individual and combined effects. M1 retains the basic backbone without multi-scale DSConv or RAA, M2 adds multi-scale DSConv to M1, M3 adds RAA, and M4 integrates both multi-scale DSConv and RAA.

As shown in Table 16, multi-scale DSConv and RAA each improve estimation on some datasets when used in isolation, but neither yields consistent gains across all degradation scenarios. Compared with the baseline M1, M2, which adds multi-scale DSConv, reduces the combined average error by approximately 1.20% and 8.45% on CX2 and Oxford, respectively. M3, which adds RAA, reduces this error by approximately 3.18%, 11.60%, and 15.49% on CS2, CX2, and Oxford, respectively. However, M2 does not achieve lower errors on CS2 or MIT, nor does M3 on MIT. These results indicate that relying solely on local degradation information extraction or cross-cycle context modeling does not yield consistent performance gains across different battery degradation scenarios.

When multi-scale DSConv and RAA are jointly integrated, the full model M4 achieves the lowest or jointly lowest combined average error at the reported precision on all four datasets, with the complementary effect being most evident on CX2. On this dataset, M2 and M3 reduce the combined average error relative to M1 by only approximately 1.20% and 11.60%, respectively, whereas M4 reduces it from 0.0500 to 0.0225, with a reported reduction of 54.94%. Even compared with M3, which uses RAA alone, M4 further reduces the error by 49.05%. These results indicate that the marked improvement of the full model reflects the complementary effect of multi-scale local degradation information

extraction and cross-cycle context modeling rather than primarily the contribution of either module in isolation.

This result is consistent with the stage transitions and nonlinear degradation observed in the CX2 data, as illustrated by CX2_38. These trajectories contain both local variations and degradation evolution across cycles. Multi-scale DSConv captures fine-grained local degradation patterns in the health indicator sequences, while RAA establishes cross-cycle context within the input window. Their joint integration allows the model to use both fine-grained local degradation patterns and cross-cycle context, providing a more complete representation of complex degradation trajectories.

The magnitude of this complementary benefit varies across datasets. On Oxford, M4 reduces the combined average error by 16.52% relative to M1, but the further reduction relative to M3 is only 1.52%. On CS2, M3 still has a slightly lower MAE than M4, while on MIT, M1 and M4 both have a combined average error of 0.0020 at the reported precision. Thus, the advantage of the full configuration does not lie in every module independently improving every metric on every dataset, but in the more consistent overall performance across degradation scenarios achieved through complementary local degradation information extraction and cross-cycle context modeling.

The preceding comparisons show that the joint configuration of multi-scale DSConv and RAA achieves more consistent overall performance across datasets. To further examine local enhancement before attention and feature refinement after local-global fusion, RAA is retained while the combinations of DSConv-S and DSConv-L are varied. Four variants are considered: S0 retains neither convolutional module, S5 retains DSConv-S, S31 retains DSConv-L, and SFull retains both. S0 and SFull correspond to M3 and M4 in Table 16, respectively.

Table 17 further shows that the benefits of retaining either convolutional module alone vary across datasets. Compared with S0, which uses RAA alone, S5 provides a small improvement only on MIT, reducing the combined average error from 0.0026 to 0.0025, or by approximately 3.85%, while achieving no lower error on CS2, CX2, or Oxford. By contrast, S31 reduces MAE, RMSE, and MAPE on CX2, lowering the combined average error from 0.0442 to 0.0348, or by approximately 21.27%, but still does not outperform S0 on CS2 or Oxford. Thus, retaining only pre-attention local enhancement or only post-fusion feature refinement does not provide a consistent advantage across all four datasets.

When DSConv-S and DSConv-L are both retained, SFull achieves the lowest combined average error on all four datasets. On CX2, the error further decreases to 0.0225, approximately 35.34% lower than that of S31; on MIT, it is approximately 20.00% lower than that of S5, the best single-convolution configuration. This result is consistent with the stage transitions and nonlinear degradation observed in the CX2 data, as illustrated by CX2_38, and further supports the design of applying convolution at different feature-processing stages. DSConv-S precedes RAA and extracts fine-grained local degradation information from neighboring cycles, so that subsequent cross-cycle context aggregation uses locally enhanced representations. In the full model, DSConv-L follows LGFA and further refines the sequence representation after local-global fusion. The two convolutional modules thus play complementary roles at different feature-processing stages, further enriching the model’s degradation representations.

However, the magnitude of the combined benefit remains dataset-dependent. SFull provides only small improvements over S0 on CS2 and Oxford, with larger improvements on CX2 and MIT. Together, the two ablation studies show complementary effects between multi-scale DSConv and RAA and between DSConv-S and DSConv-L. DSConv-S enhances fine-grained local degradation features before cross-cycle interaction, RAA establishes cross-cycle context within the input window, and DSConv-L further refines the feature representation after local-global fusion. Their joint integration provides a more complete degradation representation for SOH estimation. The current results do not establish a one-to-one correspondence between a specific convolutional kernel size and a physical degradation mechanism.

Table 16

Ablation results on different datasets.

Dataset	Model	Backbone	Multi-scale DSConv	RAA	MAE	RMSE	MAPE	Average	Reduction
CS2	M1	✓	—	—	0.0123	0.0183	0.0166	0.0157	3.74%
	M2	✓	✓	—	0.0130	0.0184	0.0171	0.0162	6.49%
	M3	✓	—	✓	0.0119	0.0178	0.0161	0.0152	0.62%
	M4	✓	✓	✓	0.0120	0.0174	0.0160	0.0151	—
	(Proposed)	✓	✓	✓	0.0120	0.0174	0.0160	0.0151	—
CX2	M1	✓	—	—	0.0170	0.0294	0.1035	0.0500	54.94%
	M2	✓	✓	—	0.0164	0.0284	0.1033	0.0494	54.40%
	M3	✓	—	✓	0.0156	0.0270	0.0899	0.0442	49.05%
	M4	✓	✓	✓	0.0106	0.0193	0.0377	0.0225	—
	(Proposed)	✓	✓	✓	0.0106	0.0193	0.0377	0.0225	—
MIT	M1	✓	—	—	0.0019	0.0022	0.0020	0.0020	2.43%
	M2	✓	✓	—	0.0021	0.0025	0.0023	0.0023	14.16%
	M3	✓	—	✓	0.0024	0.0028	0.0026	0.0026	23.91%
	M4	✓	✓	✓	0.0018	0.0022	0.0020	0.0020	—
	(Proposed)	✓	✓	✓	0.0018	0.0022	0.0020	0.0020	—
Oxford	M1	✓	—	—	0.0063	0.0076	0.0073	0.0071	16.52%
	M2	✓	✓	—	0.0058	0.0070	0.0067	0.0065	9.50%
	M3	✓	—	✓	0.0054	0.0063	0.0062	0.0060	1.52%
	M4	✓	✓	✓	0.0053	0.0062	0.0062	0.0059	—
	(Proposed)	✓	✓	✓	0.0053	0.0062	0.0062	0.0059	—

Average denotes the arithmetic mean of MAE, RMSE, and MAPE; Reduction denotes the relative decrease in Average achieved by M4 with respect to the corresponding ablation variant.

Table 17

Effects of DSConv-S and DSConv-L configurations.

Dataset	Model	DSConv-S	DSConv-L	RAA	MAE	RMSE	MAPE	Average
CS2	S0	—	—	✓	0.0119	0.0178	0.0161	0.0152
	S5	✓	—	✓	0.0128	0.0184	0.0170	0.0161
	S31	—	✓	✓	0.0124	0.0179	0.0166	0.0156
	SFull	✓	✓	✓	0.0120	0.0174	0.0160	0.0151
CX2	S0	—	—	✓	0.0156	0.0270	0.0899	0.0442
	S5	✓	—	✓	0.0162	0.0285	0.1036	0.0494
	S31	—	✓	✓	0.0135	0.0232	0.0677	0.0348
	SFull	✓	✓	✓	0.0106	0.0193	0.0377	0.0225
MIT	S0	—	—	✓	0.0024	0.0028	0.0026	0.0026
	S5	✓	—	✓	0.0023	0.0027	0.0025	0.0025
	S31	—	✓	✓	0.0025	0.0028	0.0026	0.0026
	SFull	✓	✓	✓	0.0018	0.0022	0.0020	0.0020
Oxford	S0	—	—	✓	0.0054	0.0063	0.0062	0.0060
	S5	✓	—	✓	0.0056	0.0066	0.0065	0.0062
	S31	—	✓	✓	0.0055	0.0065	0.0064	0.0061
	SFull	✓	✓	✓	0.0053	0.0062	0.0062	0.0059

4.6.2 Model complexity analysis

In practical applications, computational efficiency and storage requirements are important evaluation criteria alongside estimation accuracy [73]. This study compares the resource consumption of MS-AgentNet, CNN-Transformer, CNN-LSTM, Transformer, and LSTM using four metrics: FLOPs, training time, trainable parameter count, and storage size. Operation counts obtained using the `profile` function in the THOP library are supplemented with counts for attention operations and then converted to the number of floating-point operations required for a single

forward pass. Training time is recorded in seconds using Python’s `time` module and represents the time needed to complete the specified number of training epochs. The total number of trainable parameters is counted using PyTorch. Storage size is measured using Python’s `os.path.getsize` function and converted to KB, representing the space required to save the model weights. All metrics are measured in the same experimental environment.

To ensure a fair comparison, all five models use HI1 and HI2 from CS2_36 as inputs, with the number of training epochs, learning rate, batch size, and dropout rate set to 1000, 0.01, 128, and 0.1, respectively. For the structural setting n , MS-AgentNet, CNN-Transformer, and Transformer use four attention heads, whereas CNN-LSTM and LSTM use four LSTM layers. The embedding and MLP hidden dimensions of the attention-based models are set to 16, and the hidden dimension of the LSTM-based models is also set to 16. The configurations and complexity results are summarized in Table 18.

As shown in Table 18, MS-AgentNet requires 0.045760 M FLOPs, has 4,643 trainable parameters, and occupies 27.44 KB of weight storage, the lowest values among the five models. Compared with the standard Transformer, it requires fewer forward-pass operations and less storage, with a similar parameter count. It also uses fewer parameters and less storage than CNN-Transformer and CNN-LSTM. Compared with LSTM, its forward-pass computation is approximately 50.5% lower.

These results are consistent with the lightweight design described above. Depthwise separable convolution reduces dense cross-channel operations while extracting local degradation features. RAA uses a small number of agents to reduce the cost of cross-cycle interactions within the input window and employs learnable channel scaling to reduce the parameters required for query, key, and value generation.

MS-AgentNet has the longest training time among the five models, at 123.338 s. Its lower forward-pass computation therefore does not translate into faster training under the tested conditions.

To examine storage overhead as model width changes, the representation or hidden dimension of each model is set to 16, 32, 64, and 128, with all other settings unchanged. As shown in Fig. 10, storage size increases with model width for all five models, but MS-AgentNet maintains the lowest value at all four tested widths. At a dimension of 128, its storage size is 714.37 KB, 32.2% lower than that of Transformer, the closest model, at the same dimension. This storage advantage is maintained across the tested model widths and is not limited to the default configuration.

Together with the SOH estimation results, these comparisons indicate a favorable balance between estimation accuracy and resource overhead. Lower forward-pass computation and a smaller weight storage footprint help address the computational and storage constraints of resource-constrained BMS. These results support the potential of MS-AgentNet for lightweight SOH estimation, while online inference latency and runtime memory usage remain to be evaluated on embedded hardware.

Table 18

Model complexity comparison under standardized settings.

Models	Training hyperparameters			Model hyperparameters			Performance indicators			
	e	lr	b	d_e	d_d	n	FLOPs (M)	Training time (s)	Parameters	Storage size (KB)
MS-AgentNet	1000	0.01	128	16	16	4	0.045760	123.338	4643	27.44
Transformer	1000	0.01	128	16	16	4	0.049760	80.638	4673	29.60
CNN-Transformer	1000	0.01	128	16	16	4	0.053024	90.195	6241	36.97
CNN-LSTM	1000	0.01	128	–	16	4	0.067520	48.828	15912	68.33
LSTM	1000	0.01	128	–	16	4	0.092512	44.568	8769	39.90

Note: e , lr , b , and d_e denote the number of training epochs, learning rate, batch size, and embedding dimension, respectively. d_d denotes the MLP hidden dimension or LSTM hidden-state dimension, and n denotes the number of attention heads or LSTM layers, depending on the model. The lowest value for each resource metric is shown in bold.

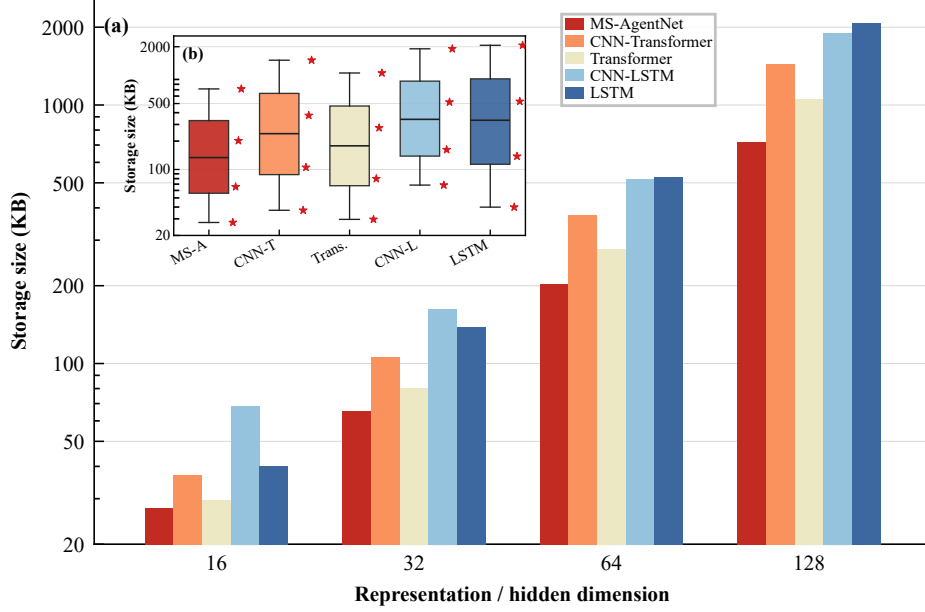


Fig. 10. Storage sizes of five models at different model widths: (a) storage size at each width; (b) overall distribution across the tested widths.

5. Conclusion

This study proposes a lightweight lithium-ion battery SOH estimation framework for resource-constrained battery management systems to address limited cross-cell HI stability and the difficulty of balancing prediction accuracy with computational efficiency. The study focuses on systematic HI construction and lightweight network design. The proposed multi-source health indicator extraction and optimization algorithm first constructs multiple types of candidate HIs from charging and discharging data and their derived curves. Based on correlations on the feature-development cell set, MS-CCCT then adaptively calibrates the constant-current charging voltage window at multiple scales, and PCC/SCC dual-threshold admission and redundancy removal determine the model inputs. With feature definitions and parameters held fixed, the selected HIs retain strong linear and monotonic relationships with SOH on other cells within the same dataset. For sequence modeling, MS-AgentNet combines ReLU² agent attention with small- and large-kernel depthwise separable convolutions to enhance the representation of local degradation variations and degradation trends over longer time scales through local feature extraction and global context modeling. With a fixed number of agents, RAA uses a small number of static learnable agents for information aggregation and broadcasting, reducing the theoretical complexity of attention-based correlation interactions from $O(N^2d)$ to $O(Nn_ad)$.

Experiments on four public battery datasets, Oxford, CALCE CS2, CALCE CX2, and MIT/Severson, show that MS-AgentNet achieves better overall performance than the four baseline models, CNN-Transformer, CNN-LSTM, Transformer, and LSTM. Specifically, the average MAPE of MS-AgentNet over the two cells in the CALCE CX2 dataset is 52.69% lower than that of CNN-Transformer. Under the standardized complexity-test settings, MS-AgentNet also has low forward-pass computation, a small parameter count, and a small weight storage footprint. Ablation studies further show complementary effects between multi-scale DSConv and RAA and between DSConv-S and DSConv-L, contributing to more consistent overall performance across degradation scenarios. Overall, the proposed framework achieves a good balance between SOH estimation accuracy and resource consumption. It maintains good estimation performance on the test cells in the corresponding datasets, showing a degree of in-domain cross-cell generalization and potential for application in resource-constrained battery management systems.

Despite these results, the framework still relies on identifiable charging and discharging data segments, and its cross-dataset adaptation performance is affected by domain differences, source-domain selection, and the amount of target-domain data. Future work will focus on HI construction for incomplete charging and dynamic operating segments, together with lightweight domain adaptation strategies. Further validation will cover more battery chemistries,

real-vehicle operating data, and embedded hardware platforms to improve the framework's practical applicability and deployment reliability.

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